6

THEORIES OF PRESHEAF TYPE: GENERAL CRITERIA

In this chapter we establish a characterization theorem providing necessary and sufficient semantic conditions for a theory to be of presheaf type (i.e. classified by a presheaf topos). This theorem subsumes the partial results previously obtained on the subject and has several corollaries which can be used in practice for testing whether a given theory is of presheaf type as well as for generating new examples of theories belonging to this class. Along the way we obtain a number of other results of independent interest, including methods for constructing theories classified by a given presheaf topos.

The contents of the chapter can be summarized as follows.

In section 6.1, in order to set up the field for the statement and proof of the characterization theorem, we identify some notable properties of theories of presheaf type. More specifically, we show that every finitely presentable model of such a theory is finitely presented, in a strong sense which we make precise in section 6.2.2, and admits an entirely syntactic description in terms of the signature of the theory and the notion of provability of geometric sequents in it (cf. section 6.1.5). We also give a semantic description of universal models of theories of presheaf type, from which we deduce a definability theorem, and establish a syntactic criterion for a theory to be classified by a presheaf topos.

In section 6.2, we show that for any geometric theory $\mathbb{T}$ and any Grothendieck topos $\mathcal{E}$ there exists, for each pair of $\mathbb{T}$-models M and N in $\mathcal{E}$, an 'object of $\mathbb{T}$-model homomorphisms' in $\mathcal{E}$ from M to N which classifies the $\mathbb{T}$-model homomorphisms between 'localizations' of M and N in slices of $\mathcal{E}$ (cf. section 6.2.1). Moreover, we introduce two notions of internal finite presentability for models of geometric theories in Grothendieck toposes, and show that the 'constant' models of a theory of presheaf type associated with its finitely presentable (set-based) models satisfy both of them.

In section 6.3, we establish our main characterization theorem providing necessary and sufficient conditions for a geometric theory to be of presheaf type. We first state the result abstractly and then proceed to obtain explicit reformulations of each of its conditions. We also derive some corollaries which allow us to verify their satisfaction in specific situations which naturally arise in practice. Lastly we show that, once expressed in the language of indexed colimits of internal diagrams in toposes, the conditions of the characterization theorem for a

Theories, Sites, Toposes. Olivia Caramello.

given geometric theory $\mathbb{T}$ amount precisely to the requirement that every model of $\mathbb{T}$ in any Grothendieck topos $\mathcal{E}$ should be a canonical $\mathcal{E}$-indexed colimit of a certain $\mathcal{E}$-filtered diagram of 'constant' models of $\mathbb{T}$ which are $\mathcal{E}$-finitely presentable.

6.1 Preliminary results

In this section we establish some results on theories of presheaf type which will be important in the sequel.

6.1.1 *A canonical form for Morita equivalences*

By Diaconescu's equivalence, any theory of presheaf type $\mathbb{T}$ classified by a topos $[\mathcal{C}, \mathbf{Set}]$ comes equipped with a Morita equivalence

$$\xi^{\mathcal{E}} : \mathbf{Flat}(\hat{\mathcal{C}}^{\mathrm{op}}, \mathcal{E}) \xrightarrow{\sim} \mathbb{T}\text{-mod}(\mathcal{E})$$

with the theory of flat functors on the category $\hat{\mathcal{C}}^{\mathrm{op}}$, where $\hat{\mathcal{C}}$ is the Cauchy completion of $\mathcal{C}$ (cf. section 5.3 above). Since $\hat{\mathcal{C}}$ can be recovered from Ind-$\hat{\mathcal{C}} \simeq \mathbf{Flat}(\hat{\mathcal{C}}^{\mathrm{op}}, \mathbf{Set})$ as the full subcategory on the finitely presentable objects (cf. Proposition C4.2.2 [55]), we can suppose $\hat{\mathcal{C}} = \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ without loss of generality, where f.p.$\mathbb{T}$-mod($\mathbf{Set}$) is the full subcategory of $\mathbb{T}$-mod($\mathbf{Set}$) on the finitely presentable objects (cf. section 6.1.4 below – this notation actually agrees with that of section 2.1.5 by Remark 6.1.14). Notice that, whilst f.p.$\mathbb{T}$-mod($\mathbf{Set}$) is not small in general, it is always essentially small (i.e. equivalent to a small category) by Lemma D2.3.2(i) [55]. So from now on, to avoid any size issues, we shall denote by f.p.$\mathbb{T}$-mod($\mathbf{Set}$) a *skeleton* of the category of finitely presentable $\mathbb{T}$-models.

Via the equivalence $\xi^{\mathcal{E}}$, the functor $y^{\mathcal{E}}_{\hat{\mathcal{C}}} : \hat{\mathcal{C}} \to \mathbf{Flat}(\hat{\mathcal{C}}^{\mathrm{op}}, \mathcal{E})$ considered in section 5.3 corresponds to the functor from f.p.$\mathbb{T}$-mod($\mathbf{Set}$) to $\mathbb{T}$-mod($\mathcal{E}$) given by $\gamma^{*}_{\mathcal{E}}$; in particular, the equivalence $\xi^{\mathbf{Set}} : \mathbf{Flat}(\hat{\mathcal{C}}^{\mathrm{op}}, \mathbf{Set}) \simeq \mathbb{T}\text{-mod}(\mathbf{Set})$ restricts to a natural equivalence

$$\tau_{\xi} : \hat{\mathcal{C}} \xrightarrow{\sim} \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}),$$

as in the following diagram:

$$\begin{tikzcd}
\hat{\mathcal{C}} \arrow[r, "\sim", "\tau_{\xi}"'] \arrow[d, "y^{\mathbf{Set}}_{\hat{\mathcal{C}}}"'] & \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}) \arrow[d, "i"] \\
\mathbf{Flat}(\hat{\mathcal{C}}^{\mathrm{op}}, \mathbf{Set}) \arrow[r, "\sim"] & \mathbb{T}\text{-mod}(\mathbf{Set}).
\end{tikzcd}$$

It is important to note that if $\hat{\mathcal{C}} = \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$, we can modify ξ so that $\tau_{\xi}(c) \cong c$ naturally in $c \in \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$. Indeed, it suffices to compose ξ with the equivalences

$$\mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}(\hat{\mathcal{C}}^{\mathrm{op}}, \mathcal{E})$$

(natural in $\mathcal{E} \in \mathfrak{Btop}$) induced by composition with $({\tau_{\xi}}^{-1})^{\mathrm{op}}$.

We can thus assume, given a theory of presheaf type $\mathbb{T}$, that $\mathbb{T}$ comes equipped with a Morita equivalence ξ satisfying the condition $\tau_\xi \cong 1_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}$; we call such an equivalence *canonical*, and accordingly we say that an equivalence

$$\chi^{\mathcal{E}} : \mathbb{T}\text{-mod}(\mathcal{E}) \simeq \mathbf{Geom}(\mathcal{E}, [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}])$$

natural in $\mathcal{E} \in \mathfrak{Btop}$ is canonical if it is induced by a canonical equivalence

$$\mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E}) \xrightarrow{\sim} \mathbb{T}\text{-mod}(\mathcal{E})$$

by composition with Diaconescu's equivalence.

Note that 'the' canonical Morita equivalence for a theory of presheaf type $\mathbb{T}$ is induced by the canonical geometric morphism

$$[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

via Diaconescu's equivalence. This yields in particular a canonical representation for the classifying topos of a theory of presheaf type $\mathbb{T}$ as the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$. Notice that, since

$$\mathbb{T}\text{-mod}(\mathbf{Set}) \simeq \mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathbf{Set}),$$

the category $\mathbb{T}$-mod($\mathbf{Set}$) is the ind-completion of the category f.p.$\mathbb{T}$-mod($\mathbf{Set}$) (cf. section 8.1.3 below for some general results about ind-completions).

6.1.2 *Universal models and definability*

The following result gives an explicit description of 'the' universal model of a theory of presheaf type.

Theorem 6.1.1 (Theorem 3.1 [19]). *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ. Then the Σ-structure $N_{\mathbb{T}}$ in* $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ *which assigns to a sort A of Σ the evaluation functor $N_{\mathbb{T}}A$ at A (sending any model $M \in$* f.p.$\mathbb{T}$-mod($\mathbf{Set}$) *to the set MA and acting on the arrows accordingly), to a function symbol $f : A_1 \cdots A_n \to B$ of Σ the morphism $N_{\mathbb{T}}A_1 \times \cdots \times N_{\mathbb{T}}A_n \to N_{\mathbb{T}}B$ given by $(N_{\mathbb{T}}f)(M) = Mf$ and to a relation symbol $R \rightarrowtail A_1 \cdots A_n$ of Σ the subobject $N_{\mathbb{T}}R \rightarrowtail N_{\mathbb{T}}A_1 \times \cdots \times N_{\mathbb{T}}A_n$ given by $(N_{\mathbb{T}}R)(M) = MR$ (for any $M \in$* f.p.$\mathbb{T}$-mod($\mathbf{Set}$)*) is a universal model for $\mathbb{T}$; moreover, for any geometric formula $\phi(\vec{x})$ over Σ, the interpretation $[[\vec{x} \,.\, \phi]]_{N_{\mathbb{T}}}$ of $\phi(\vec{x})$ in $N_{\mathbb{T}}$ is given by $[[\vec{x} \,.\, \phi]]_{N_{\mathbb{T}}}(M) = [[\vec{x} \,.\, \phi]]_M$ for any $M \in$* f.p.$\mathbb{T}$-mod($\mathbf{Set}$). *In particular, the finitely presentable $\mathbb{T}$-models are jointly conservative for $\mathbb{T}$.*

Proof Consider a canonical Morita equivalence

$$\xi^{\mathcal{E}} : \mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E}) \xrightarrow{\sim} \mathbb{T}\text{-mod}(\mathcal{E})$$

for the theory of presheaf type $\mathbb{T}$. By composing it with Diaconescu's equivalence

$$\mathbf{Geom}(\mathcal{E}, [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]) \simeq \mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathcal{E}),$$

we get an equivalence $\tau^{\mathcal{E}} : \mathbf{Geom}(\mathcal{E}, [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$ natural in $\mathcal{E} \in \mathfrak{Btop}$; let us define U as the image of the identical geometric morphism on [f.p.$\mathbb{T}$-mod(**Set**), **Set**] under this equivalence. Then U is a universal model of $\mathbb{T}$ and a geometric morphism $f : \mathcal{E} \to$ [f.p.$\mathbb{T}$-mod(**Set**), **Set**] is sent by $\tau^{\mathcal{E}}$ to the $\mathbb{T}$-model $f^*(U)$. Recall that, via Diaconescu's equivalence, the identity on [f.p.$\mathbb{T}$-mod(**Set**), **Set**] is sent to the flat functor y : f.p.$\mathbb{T}$-mod(**Set**)$^{\text{op}} \to$ [f.p.$\mathbb{T}$-mod(**Set**), **Set**] given by the Yoneda embedding. So, by naturality, for any model $M \in$ f.p.$\mathbb{T}$-mod(**Set**), the geometric morphism

$$e_M : \mathbf{Set} \to [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

whose inverse image is the evaluation functor ev_M : [f.p.$\mathbb{T}$-mod(**Set**), **Set**] $\to$ **Set** at M is sent to the flat functor $(e_M)^* \circ y = \text{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(-, M)$: f.p.$\mathbb{T}$-mod(**Set**)$^{\text{op}} \to$ **Set**. But, since ξ is canonical, $\xi^{\mathbf{Set}}$ sends the functor $\text{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(-, M)$ to the model M and hence $\tau^{\mathbf{Set}}$ sends e_M to M (for any $M \in$ f.p.$\mathbb{T}$-mod(**Set**)). Thus $M \cong (e_M)^*(U) = \text{ev}_M(U)$ for any $M \in$ f.p.$\mathbb{T}$-mod(**Set**), and hence U is (isomorphic to) the Σ-structure $N_{\mathbb{T}}$ defined in the statement of the theorem. Now, the fact that $[[\vec{x} \,.\, \phi]]_{N_{\mathbb{T}}}(M) = [[\vec{x} \,.\, \phi]]_M$ for any geometric formula $\phi(\vec{x})$ and model $M \in$ f.p.$\mathbb{T}$-mod(**Set**) follows from the fact that the functors ev_M are geometric (being inverse image functors of geometric morphisms). □

Remarks 6.1.2 (a) Theorem 6.1.1 specializes to Corollary D3.1.2 [55] in the case when $\mathbb{T}$ is cartesian.

(b) Since universal models are preserved by inverse image functors of geometric inclusions (Lemma 2.1 [19]), 'the' universal model of a quotient $\mathbb{T}'$ of a theory of presheaf type $\mathbb{T}$ is given by the image of the model $N_{\mathbb{T}}$ under the associated sheaf functor $\mathcal{E}_{\mathbb{T}} \to \mathcal{E}_{\mathbb{T}'}$. In particular, if for each sort A of the signature of $\mathbb{T}$ the formula $\{x^A \,.\, \top\}$ presents a $\mathbb{T}$-model then the functors $N_{\mathbb{T}}A$ are all representable and hence, if the induced $\mathbb{T}$-topology J of $\mathbb{T}$ (in the sense of section 8.1.2) is subcanonical, the assignment $A \to N_{\mathbb{T}}A$ yields a universal model of $\mathbb{T}'$ inside its classifying topos $\mathbf{Sh}$(f.p.$\mathbb{T}$-mod(**Set**)$^{\text{op}}, J) \hookrightarrow$ [f.p.$\mathbb{T}$-mod(**Set**), **Set**].

(c) The proof of Theorem 6.1.1 shows that if $\mathbb{T}$ is a geometric theory classified by a topos $[\mathcal{C}, \mathbf{Set}]$ then the $\mathbb{T}$-models corresponding to the objects of $\mathcal{C}$ are jointly conservative for $\mathbb{T}$.

As shown by the following theorem, universal models of geometric theories enjoy a strong form of logical completeness.

Theorem 6.1.3 (Theorem 2.2 [19]). *Let $\mathbb{T}$ be a geometric theory over a signature Σ and U a universal model of $\mathbb{T}$ in a Grothendieck topos $\mathcal{E}$. Then*

(i) *For any subobject $S \rightarrowtail UA_1 \times \cdots \times UA_n$ in $\mathcal{E}$, there exists a geometric formula $\phi(\vec{x}) = \phi(x_1^{A_1}, \ldots, x_n^{A_n})$ over Σ such that $S = [[\vec{x} \,.\, \phi]]_U$.*

(ii) *For any arrow $f : [[\vec{x} \,.\, \phi]]_U \to [[\vec{y} \,.\, \psi]]_U$ in $\mathcal{E}$, where $\phi(\vec{x})$ and $\psi(\vec{y})$ are geometric formulae over Σ, there exists a geometric formula $\theta(\vec{x}, \vec{y})$ over Σ such that the sequents $(\phi \vdash_{\vec{x}} (\exists \vec{y})\theta)$, $(\theta \vdash_{\vec{x},\vec{y}} \phi \wedge \psi)$ and $(\theta \wedge \theta[\vec{y'}/\vec{y}] \vdash_{\vec{x},\vec{y},\vec{y'}} \vec{y} = \vec{y'})$ are provable in $\mathbb{T}$ and $[[\vec{x}, \vec{y} \,.\, \theta]]_U$ is the graph of f.*

Proof We can clearly suppose, without loss of generality, that U is the universal model $U_{\mathbb{T}}$ of $\mathbb{T}$ lying in its classifying topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ described in Remark 2.1.9.

To prove (i) we observe that, given a geometric formula $\phi(x_1^{A_1}, \ldots, x_n^{A_n})$ over Σ, its interpretation in $U_{\mathbb{T}}$ identifies with the sieve on $\{x_1^{A_1}, \ldots, x_n^{A_n} \,.\, \top\}$ generated by the canonical monomorphism $[\phi] : \{\vec{x} \,.\, \phi\} \rightarrowtail \{x_1^{A_1}, \ldots, x_n^{A_n} \,.\, \top\}$. Now, if S is a subobject of

$$U_{\mathbb{T}}A_1 \times \cdots \times U_{\mathbb{T}}A_n \cong \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(-, \{x_1^{A_1}, \ldots, x_n^{A_n} \,.\, \top\})$$

then S, regarded as a sieve, is $J_{\mathbb{T}}$-closed and hence, by Proposition 3.1.6(ii), it is generated by a subobject of $\{x_1^{A_1}, \ldots, x_n^{A_n} \,.\, \top\}$ in $\mathcal{C}_{\mathbb{T}}$. Thus (i) follows from the characterization of subobjects in $\mathcal{C}_{\mathbb{T}}$ given by Lemma 1.4.2.

Let us now prove (ii). By the Yoneda lemma, any arrow

$$f : [[\vec{x} \,.\, \phi]]_{U_{\mathbb{T}}} \cong \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(-, \{\vec{x} \,.\, \phi\}) \to \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(-, \{\vec{y} \,.\, \psi\}) \cong [[\vec{y} \,.\, \psi]]_{U_{\mathbb{T}}}$$

in $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is of the form $\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(-, [\theta])$ for a unique arrow $[\theta] : \{\vec{x} \,.\, \phi\} \to \{\vec{y} \,.\, \psi\}$ in $\mathcal{C}_{\mathbb{T}}$; but $[[\vec{x}, \vec{y} \,.\, \theta]]_{U_{\mathbb{T}}}$ is the graph of the arrow $\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(-, [\theta])$, from which our thesis follows. □

Theorem 6.1.3, combined with Theorem 6.1.1, yields the following definability theorem for theories of presheaf type.

Theorem 6.1.4 (Corollary 3.2 [19]). *Let $\mathbb{T}$ be a theory of presheaf type and $A_1, \ldots, A_n$ sorts of its signature. Suppose that we are given, for every finitely presentable set-based model M of $\mathbb{T}$, a subset R_M of $MA_1 \times \cdots \times MA_n$ in such a way that every $\mathbb{T}$-model homomorphism $h : M \to N$ maps R_M into R_N. Then there exists a (unique up to $\mathbb{T}$-provable equivalence) geometric formula-in-context $\phi(\vec{x})$, where $\vec{x} = (x_1^{A_1}, \ldots, x_n^{A_n})$, such that $R_M = [[\vec{x} \,.\, \phi]]_M$ for every finitely presentable $\mathbb{T}$-model M.*

Proof The thesis immediately follows from Theorems 6.1.3 and 6.1.1 observing that the assignment $M \to R_M$ in the statement of the theorem gives rise to a subfunctor $R \rightarrowtail N_{\mathbb{T}}$, where $N_{\mathbb{T}}$ is the universal model of $\mathbb{T}$ in [f.p.$\mathbb{T}$-mod($\mathbf{Set}$), $\mathbf{Set}$] defined in the statement of Theorem 6.1.1. Conversely, any geometric formula $\phi(\vec{x})$ yields a covariant functorial assignment $M \to [[\vec{x} \,.\, \phi]]_M$, since any $\mathbb{T}$-model homomorphism $h : M \to N$ maps $[[\vec{x} \,.\, \phi]]_M$ to $[[\vec{x} \,.\, \phi]]_N$ (cf. Remark 1.3.21). In other words, we have the following 'bridge':

Subobject of $UA_1 \times \cdots \times UA_n$
$[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$

f.p.$\mathbb{T}$-mod($\mathbf{Set}$)$^{\mathrm{op}}$
Functorial (covariant) assignment
$M \to R_M \subseteq MA_1 \times \cdots \times MA_n$

$(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$
Geometric formula
$\phi(x_1^{A_1}, \ldots, x_n^{A_n})$

where U is 'the' universal model of $\mathbb{T}$ in its classifying topos. □

Remarks 6.1.5 (a) The proof of Theorem 6.1.4 also shows that, for any two geometric formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ over the signature of $\mathbb{T}$, every assignment $M \to f_M : [[\vec{x} \,.\, \phi]]_M \to [[\vec{y} \,.\, \psi]]_M$ (for finitely presentable $\mathbb{T}$-models M) which is natural in M is definable in finitely presentable $\mathbb{T}$-models by a $\mathbb{T}$-provably functional formula $\theta(\vec{x}, \vec{y})$ from $\phi(\vec{x})$ to $\psi(\vec{y})$.

(b) If the property R of tuples $\vec{x}$ of elements of finitely presentable $\mathbb{T}$-models in the statement of Theorem 6.1.4 is extended to all set-based $\mathbb{T}$-models so as to be preserved by filtered colimits in $\mathbb{T}$-mod(**Set**) then we have $R_M = [[\vec{x} \,.\, \phi]]_M$ for each set-based $\mathbb{T}$-model M.

(c) If in Theorem 6.1.4 the theory $\mathbb{T}$ is coherent and the property R is not only preserved but also reflected by arbitrary $\mathbb{T}$-model homomorphisms then the formula $\phi(\vec{x})$ in the statement of the theorem can be taken to be coherent and $\mathbb{T}$-Boolean (in the sense that there exists a coherent formula $\psi(\vec{x})$ in the same context such that the sequents $(\phi \wedge \psi \vdash_{\vec{x}} \bot)$ and $(\top \vdash_{\vec{x}} \phi \vee \psi)$ are provable in $\mathbb{T}$). Indeed, the theorem can be applied both to the property R and to the negation of it yielding two geometric formulae $\phi(\vec{x})$ and $\psi(\vec{x})$ defining them such that $(\phi \wedge \psi \vdash_{\vec{x}} \bot)$ and $(\top \vdash_{\vec{x}} \phi \vee \psi)$ are provable in $\mathbb{T}$. Since $\mathbb{T}$ is coherent and every geometric formula in a given context is provably equivalent to a disjunction of coherent formulae in the same context, we can suppose ϕ and ψ to be coherent without loss of generality (cf. Theorem 10.8.6(iii)).

6.1.3 *A syntactic criterion for a theory to be of presheaf type*

In this section we shall establish a syntactic criterion for a geometric theory to be of presheaf type.

The following notion will be central in our analysis:

Definition 6.1.6 (Definition C2.2.18 [55]) A Grothendieck topology J on a category $\mathcal{C}$ is said to be *rigid* if, for every object c of $\mathcal{C}$, the collection of arrows from J-irreducible objects of $\mathcal{C}$ (i.e. objects whose J-covering sieves are only the maximal ones) generates a J-covering sieve.

In Theorem 6.1.7 below, for a subcanonical site $(\mathcal{C}, J)$, we denote by $y : \mathcal{C} \to \mathbf{Sh}(\mathcal{C}, J)$ the factorization through $\mathbf{Sh}(\mathcal{C}, J) \hookrightarrow [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ of the Yoneda embedding.

Theorem 6.1.7 (Theorem 3.13 [20]). *Let $(\mathcal{C}, J)$ be a small subcanonical site such that $y(\mathcal{C})$ is closed in $\mathbf{Sh}(\mathcal{C}, J)$ under retracts. Then $\mathbf{Sh}(\mathcal{C}, J)$ is equivalent to a presheaf topos if and only if J is rigid.*

Proof The 'if' direction follows at once from the Comparison Lemma (Theorem 1.1.8). It thus remains to prove the 'only if' direction. If $\mathcal{E} = \mathbf{Sh}(\mathcal{C}, J)$ is equivalent to a presheaf topos then, by Lemma C2.2.20 [55], $\mathcal{E}$ has a separating set of indecomposable projective objects. Now, suppose A is an indecomposable projective object of $\mathcal{E}$. Then, as it is observed in the proof of Lemma C2.2.20 [55], given any epimorphic family $\{f_i : B_i \to A \mid i \in I\}$, at least one f_i must be a split epimorphism; in particular, A is $J_{\mathcal{E}}$-irreducible, where $J_{\mathcal{E}}$ is the canonical topology on $\mathcal{E}$. Hence, by taking as epimorphic family the collection of all the arrows in $\mathcal{E}$ from objects of the form $y(c)$ to A, we obtain that A is a retract in $\mathcal{E}$ of an object of the form $y(c)$ for some $c \in \mathcal{C}$. Thus, by our hypotheses, A is itself, up to isomorphism, of the form $y(c)$. Let us denote by $\mathcal{C}'$ the full subcategory of $\mathcal{C}$ on the objects c such that $y(c)$ is indecomposable and projective in $\mathcal{E}$; then the objects in $y(\mathcal{C}')$ form a separating set of objects for $\mathcal{E}$. Thus, for any object B of $\mathcal{E}$, the family of all the arrows in $\mathcal{E}$ from objects of the form $y(c)$, for $c \in \mathcal{C}'$, to B generates a $J_{\mathcal{E}}$-covering sieve. Now, J being subcanonical, $J = J_{\mathcal{E}}|_{\mathcal{C}}$ (by Proposition C2.2.16 [55]) and hence, for any object $c \in \mathcal{C}$, the collection of all arrows in $\mathcal{C}$ from objects of $\mathcal{C}'$ to c is J-covering; but every J-covering sieve on an object of $\mathcal{C}'$ is trivial, so J is rigid, as required. $\square$

Remark 6.1.8 Under the hypotheses of the theorem, if $\mathcal{C}$ is Cauchy-complete (in particular, if $\mathcal{C}$ is cartesian) then $y(\mathcal{C})$ is closed in $\mathcal{E} = \mathbf{Sh}(\mathcal{C}, J)$ under retracts. Indeed, let $i : A \rightarrowtail y(c)$, $r : y(c) \twoheadrightarrow A$ be a retract of A in $\mathcal{E}$, that is, $r \circ i = 1_A$. Then $i \circ r : y(c) \to y(c)$ is idempotent. Now, since y is full and faithful, $i \circ r = y(e)$ for some idempotent $e : c \to c$ in $\mathcal{C}$. Since $\mathcal{C}$ is Cauchy-complete, e splits as $s \circ t$ where $t \circ s = 1$. Then $y(s)$ and $y(t)$ form a retract of A and hence, by the uniqueness up to isomorphism of the splitting of an idempotent in a category, r is isomorphic to $y(t)$ and i is isomorphic to $y(s)$; in particular, A is isomorphic to $y(\mathrm{dom}(s))$.

By Remark 1.4.9, the regular (resp. coherent, geometric) syntactic sites of regular (resp. coherent, geometric) theories all satisfy the hypotheses of Theorem 6.1.7. This motivates the following definition:

Definition 6.1.9 Let $\mathbb{T}$ be a geometric theory over a signature Σ. A geometric formula-in-context $\{\vec{x} \,.\, \phi\}$ is said to be $\mathbb{T}$-*irreducible* if it is $J_{\mathbb{T}}$-irreducible as an object of the syntactic category $\mathcal{C}_{\mathbb{T}}$, equivalently if for any family $\{\theta_i \mid i \in I\}$ of $\mathbb{T}$-provably functional geometric formulae $\{\vec{x_i}, \vec{x} \,.\, \theta_i\}$ from $\{\vec{x_i} \,.\, \phi_i\}$ to $\{\vec{x} \,.\, \phi\}$ such that $(\phi \vdash_{\vec{x}} \bigvee\limits_{i \in I} (\exists \vec{x_i})\theta_i)$ is provable in $\mathbb{T}$, there exist $i \in I$ and a $\mathbb{T}$-provably functional geometric formula $\{\vec{x}, \vec{x_i} \,.\, \theta'\}$ from $\{\vec{x} \,.\, \phi\}$ to $\{\vec{x_i} \,.\, \phi_i\}$ such that the composite arrow $[\theta_i] \circ [\theta']$ in $\mathcal{C}_{\mathbb{T}}$ is equal to the identity on $\{\vec{x} \,.\, \phi\}$ (equivalently, the bisequent $(\phi(\vec{x}) \wedge \vec{x} = \vec{x'} \dashv\vdash_{\vec{x}, \vec{x'}} (\exists \vec{x_i})(\theta'(\vec{x'}, \vec{x_i}) \wedge \theta_i(\vec{x_i}, \vec{x})))$ is provable in $\mathbb{T}$).

Theorem 6.1.7 thus yields the following syntactic criterion for a geometric theory to be of presheaf type (cf. also the proof of Theorem 10.3.4(i)):

Corollary 6.1.10 (Corollary 3.15 [20]). *Let $\mathbb{T}$ be a geometric theory over a signature Σ. Then $\mathbb{T}$ is of presheaf type if and only if every geometric formula-in-context over Σ, regarded as an object of $\mathcal{C}_{\mathbb{T}}$, is $J_{\mathbb{T}}$-covered by $\mathbb{T}$-irreducible formulae, i.e. if and only if there exists a collection $\mathcal{F}$ of geometric formulae-in-context over Σ satisfying the following properties:*

(i) *For any geometric formula* $\{\vec{y}\,.\,\psi\}$ *over* Σ, *there exist objects* $\{\vec{x_i}\,.\,\phi_i\}$ *in* $\mathcal{F}$ *(for* $i \in I$*) and* $\mathbb{T}$*-provably functional geometric formulae* $\{\vec{x_i}, \vec{y}\,.\,\theta_i\}$ *from* $\{\vec{x_i}\,.\,\phi_i\}$ *to* $\{\vec{y}\,.\,\psi\}$ *such that* $(\psi \vdash_{\vec{y}} \bigvee_{i\in I}(\exists \vec{x_i})\theta_i)$ *is provable in* $\mathbb{T}$.

(ii) *Every formula* $\{\vec{x}\,.\,\phi\}$ *in* $\mathcal{F}$ *is* $\mathbb{T}$*-irreducible, i.e. for any family* $\{\theta_i \mid i \in I\}$ *of* $\mathbb{T}$*-provably functional geometric formulae* $\{\vec{x_i}, \vec{x}\,.\,\theta_i\}$ *from* $\{\vec{x_i}\,.\,\phi_i\}$ *to* $\{\vec{x}\,.\,\phi\}$ *such that* $(\phi \vdash_{\vec{x}} \bigvee_{i\in I}(\exists \vec{x_i})\theta_i)$ *is provable in* $\mathbb{T}$, *there exist* $i \in I$ *and a* $\mathbb{T}$*-provably functional geometric formula* $\{\vec{x}, \vec{x_i}\,.\,\theta'\}$ *from* $\{\vec{x}\,.\,\phi\}$ *to* $\{\vec{x_i}\,.\,\phi_i\}$ *such that the bisequent* $(\phi(\vec{x}) \wedge \vec{x} = \vec{x'} \dashv\vdash_{\vec{x},\vec{x'}} (\exists \vec{x_i})(\theta'(\vec{x'}, \vec{x_i}) \wedge \theta_i(\vec{x_i}, \vec{x})))$ *is provable in* $\mathbb{T}$.

6.1.4 *Finitely presentable = finitely presented*

We shall prove in this section that the semantic and syntactic notions of finite presentability for models of geometric theories coincide for all theories of presheaf type. Moreover, we shall characterize in a syntactic way the geometric formulae which present some model of a given theory of presheaf type. This generalizes the well-known result for cartesian theories (cf. section D2.4 of [55]).

First, let us recall the standard notions of finite presentability of a set-based model of a geometric theory.

Let Σ be a first-order signature. It is well known that the category Σ-str(**Set**) of Σ-structures in **Set** has all filtered colimits, which can be constructed as follows. For any diagram $D : \mathcal{I} \to \Sigma\text{-str}(\mathbf{Set})$ defined on a filtered category $\mathcal{I}$, its colimit $\mathrm{colim}(D)$ can be realized by taking, for any sort A of Σ, $\mathrm{colim}(D)A$ equal to the colimit of the functor $D_A : \mathcal{I} \to \mathbf{Set}$ given by the composite of D with the forgetful functor $\Sigma\text{-str}(\mathbf{Set}) \to \mathbf{Set}$ corresponding to the sort A. By Proposition 2.13.3 [8], this can be identified with the disjoint union $\coprod_{i\in\mathcal{I}} D_A(i)$ modulo the equivalence relation $\approx_A$ defined as follows: given $x \in D_A(i)$ and $x' \in D_A(i')$, $x \approx_A x'$ if and only if there exist arrows $f : i \to i''$ and $g : i' \to i''$ in $\mathcal{I}$ such that $(D_A(f))(x) = (D_A(g))(x')$. Let us denote by $J_i : D(i) \to \mathrm{colim}(D)$ the canonical colimit maps (for $i \in \mathcal{I}$). For any function symbol $f : A_1 \cdots A_n \to B$ in Σ, we define $\mathrm{colim}(D)f : \mathrm{colim}(D)A_1 \times \cdots \times \mathrm{colim}(D)A_n \to \mathrm{colim}(D)B$ as follows. For any $(x_1, \ldots, x_n) \in \mathrm{colim}(D)A_1 \times \cdots \times \mathrm{colim}(D)A_n$, we can suppose, thanks to the filteredness of $\mathcal{I}$, that there exist $i \in I$ and for each $k \in \{1, \ldots, n\}$ an element $x'_k \in D_{A_k}(i)$ such that $x_k = J_i(x'_k)$ for all k; we thus set $(\mathrm{colim}(D)f)((x_1, \ldots, x_n))$ equal to $J_{iB}((D(i)f)((x'_1, \ldots, x'_n)))$. It is readily seen that this map is well-defined. For any relation symbol $R \rightarrowtail A_1 \cdots A_n$ in Σ, we define $\mathrm{colim}(D)R$ as the subset of $\mathrm{colim}(D)A_1 \times \cdots \times \mathrm{colim}(D)A_n$ consisting of the tuples of the form $J_i((x_1, \ldots, x_n))$ where $(x_1, \ldots, x_n) \in D(i)R$ for some $i \in \mathcal{I}$. Notice that for any $i \in \mathcal{I}$ and any $x, x' \in D_A(i)$, $x \approx_A x'$ if and only if there exists an arrow $s : i \to j$ in $\mathcal{I}$ such that $D_A(s)(x) = D_A(s)(x')$, and that if $x \approx_A x'$ then for any arrow $t : i \to k$ in $\mathcal{I}$, $D_A(t)(x) \approx_A D_A(t)(x')$.

Recall from Lemma D2.4.9 [55] that for any geometric theory $\mathbb{T}$ over a signature Σ, the category $\mathbb{T}$-mod(**Set**) of $\mathbb{T}$-models in **Set** is closed in Σ-str(**Set**) under filtered colimits.

Definition 6.1.11 Let $\mathbb{T}$ be a geometric theory over a signature Σ and M a set-based $\mathbb{T}$-model. Then

(a) The model M is said to be *finitely presentable* if the representable functor

$$\mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(M,-) : \mathbb{T}\text{-mod}(\mathbf{Set}) \to \mathbf{Set}$$

preserves filtered colimits.

(b) The model M is said to be *finitely presented* if there is a geometric formula $\{\vec{x} \,.\, \phi\}$ over Σ and a string of elements $(\xi_1, \ldots, \xi_n) \in MA_1 \times \cdots \times MA_n$ (where $A_1, \ldots, A_n$ are the sorts of the variables in $\vec{x}$), called the *generators* of M, such that for any $\mathbb{T}$-model N in **Set** and string of elements $(b_1, \ldots, b_n) \in MA_1 \times \cdots \times MA_n$ such that $(b_1, \ldots, b_n) \in [[\vec{x} \,.\, \phi]]_N$, there exists a unique arrow $f : M \to N$ in $\mathbb{T}$-mod(**Set**) such that $(f_{A_1} \times \cdots \times f_{A_n})((\xi_1, \ldots, \xi_n)) = (b_1, \ldots, b_n)$.

The full subcategory of $\mathbb{T}$-mod(**Set**) on the finitely presentable models (or a skeleton of it) will be denoted by f.p.$\mathbb{T}$-mod(**Set**). We shall denote 'the' model finitely presented by a formula $\{\vec{x} \,.\, \phi\}$ by $M_{\{\vec{x} \,.\, \phi\}}$, and its tuple of generators by $\vec{\xi_\phi}$.

The following lemma provides an explicit categorical characterization of the finitely presentable objects of a category with filtered colimits. By the above remark, this applies in particular to the categories $\mathbb{T}$-mod(**Set**) of set-based models of a geometric theory $\mathbb{T}$; indeed, a set-based model of a geometric theory $\mathbb{T}$ is finitely presentable in the sense of Definition 6.1.11(a) if and only if it is finitely presentable as an object of the category $\mathbb{T}$-mod(**Set**).

Lemma 6.1.12 *Let $\mathcal{D}$ be a category with filtered colimits and M an object of $\mathcal{D}$. Then M is finitely presentable in $\mathcal{D}$ if and only if for any diagram $D : \mathcal{I} \to \mathcal{D}$ defined on a small filtered category $\mathcal{I}$, every arrow $M \to \mathrm{colim}(D)$ factors through one of the canonical colimit arrows $J_i : D(i) \to \mathrm{colim}(D)$ and for any arrows $f : M \to D(i)$ and $g : M \to D(j)$ in $\mathcal{D}$ such that $J_i \circ f = J_j \circ g$ there exist $k \in \mathcal{I}$ and two arrows $s : i \to k$ and $t : j \to k$ in $\mathcal{I}$ such that $D(s) \circ f = D(t) \circ g$.*

Proof By definition, M is finitely presentable in $\mathcal{D}$ if and only if the functor $\mathrm{Hom}_{\mathcal{D}}(M,-) : \mathcal{D} \to \mathbf{Set}$ preserves filtered colimits, that is for any diagram $D : \mathcal{I} \to \mathcal{D}$ defined on a small filtered category $\mathcal{I}$, the canonical map ξ from the colimit in **Set** of the composite diagram $\mathrm{Hom}_{\mathcal{D}}(M,-) \circ D$ to $\mathrm{Hom}_{\mathcal{D}}(M, \mathrm{colim}(D))$ is a bijection. The former set can be described by Proposition 2.13.3 [8] as the disjoint union of the $\mathrm{Hom}_{\mathcal{D}}(M, D(i))$ (for $i \in \mathcal{I}$) modulo the equivalence relation $\approx$ defined as follows: for any $f : M \to D(i)$ and $g : M \to D(j)$, $f \approx g$ if and only if there exists $k \in \mathcal{I}$ and two arrows $s : i \to k$ and $t : j \to k$ such that $D(s) \circ f = D(t) \circ g$. The map ξ sends the $\approx$-equivalence class of an arrow

$f : M \to D(i)$ to the composite arrow $J_i \circ f$, where J_i is the canonical colimit arrow $D(i) \to \mathrm{colim}(D)$; the first and second conditions in the statement of the lemma thus correspond respectively to the requirements that ξ be surjective and injective. □

Our argument to establish the equivalence between the syntactic and semantic notions of finite presentability for a model of a theory of presheaf type $\mathbb{T}$ is based on the concept of irreducible object of a Grothendieck topos; in fact, this is precisely the invariant that we shall 'transfer' across the two distinct representations

$$[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

of the classifying topos of $\mathbb{T}$. Recall that an object A of a Grothendieck topos $\mathcal{E}$ is said to be *irreducible* if every covering sieve on A with respect to the canonical topology on $\mathcal{E}$ is maximal, that is if any sieve in $\mathcal{E}$ containing a small epimorphic family contains the identity on A.

Theorem 6.1.13 (Theorem 4.3 [20]). *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ. Then*

(i) *Any finitely presentable $\mathbb{T}$-model in* **Set** *is presented by a $\mathbb{T}$-irreducible geometric formula $\phi(\vec{x})$ over Σ.*
(ii) *Conversely, any $\mathbb{T}$-irreducible geometric formula $\phi(\vec{x})$ over Σ presents a $\mathbb{T}$-model.*

In fact, the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}$ *is equivalent to the full subcategory $\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}$ of $\mathcal{C}_{\mathbb{T}}$ on the $\mathbb{T}$-irreducible formulae.*

Proof It is clear from the proof of Theorem 6.1.7 that a geometric formula $\{\vec{x}\,.\,\phi\}$ over the signature of $\mathbb{T}$ is $\mathbb{T}$-irreducible if and only if the object $y_{\mathbb{T}}(\{\vec{x}\,.\,\phi\})$, where $y_{\mathbb{T}}$ is the Yoneda embedding

$$y_{\mathbb{T}} : \mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}),$$

is irreducible in the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.

Since $\mathbb{T}$ is of presheaf type, we have a canonical equivalence of classifying toposes

$$\tau : \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \simeq [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

(in the sense of section 6.1.1).

If $\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}$ is the full subcategory of $\mathcal{C}_{\mathbb{T}}$ on the $\mathbb{T}$-irreducible formulae then, by Theorem 6.1.7, the Comparison Lemma yields an equivalence $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \simeq [(\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}})^{\mathrm{op}}, \mathbf{Set}]$.

If $\widehat{\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}} \hookrightarrow \mathcal{C}_{\mathbb{T}}$ is the Cauchy completion of $\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}$ (as in section 5.3.1) then $[(\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}})^{\mathrm{op}}, \mathbf{Set}] \simeq [(\widehat{\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}})^{\mathrm{op}}, \mathbf{Set}]$ and the resulting equivalence

$$[\widehat{\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}}^{\mathrm{op}}, \mathbf{Set}] \simeq [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

restricts to an equivalence

$$l : \widehat{\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}}^{\mathrm{op}} \simeq \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$$

between the full subcategories on the irreducible objects (cf. the proof of Theorem 6.1.7):

$$\begin{tikzcd}[column sep=small]
& \overset{\textit{Irreducible object}}{[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})} \arrow[dl, dash, dashed] \arrow[dr, dash, dashed] & \\
\underset{\textit{Any object}}{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}} & & \underset{\textit{formula}}{\underset{\mathbb{T}\textit{-irreducible}}{(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})}}
\end{tikzcd}$$

Now, given $\{\vec{x} . \phi\} \in \widehat{\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}}$, τ sends the functor $y_{\mathbb{T}}(\{\vec{x} . \phi\}) = [[\vec{x} . \phi]]_{U_{\mathbb{T}}}$ to $y(l(\{\vec{x} . \phi\})) = [[\vec{x} . \phi]]_{N_{\mathbb{T}}}$ (where the notation is that of section 6.1.2), whence the model $l(\{\vec{x} . \phi\})$ is finitely presented by $\{\vec{x} . \phi\}$. Thus, since all representables in presheaf toposes are irreducible objects, $\phi(\vec{x})$ is $\mathbb{T}$-irreducible. So $\widehat{\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}} \simeq \mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}$, that is, $\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}$ is Cauchy-complete. Hence l yields an equivalence

$$(\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}})^{\mathrm{op}} \simeq \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}).$$

This equivalence sends any formula $\{\vec{x} . \phi\} \in \mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}$ to the model $M_{\{\vec{x} . \phi\}}$ presented by it and any arrow $[\theta] : \{\vec{y} . \psi\} \to \{\vec{x} . \phi\}$ in $\mathcal{C}_{\mathbb{T}}^{\mathrm{irr}}$ to the arrow $M_{[\theta]} : M_{\{\vec{x} . \phi\}} \to M_{\{\vec{y} . \psi\}}$ in f.p.$\mathbb{T}$-mod($\mathbf{Set}$) defined by setting $M_{[\theta]}(\vec{\xi_{\phi}})$ equal to the unique element $z \in [[\vec{x} . \phi]]_{M_{\{\vec{y} . \psi\}}}$ such that $(\vec{\xi_{\psi}}, z) \in [[\vec{y}, \vec{x} . \theta]]_{M_{\{\vec{y} . \psi\}}}$. $\square$

Remark 6.1.14 Theorem 6.1.13 generalizes Theorem 2.1.21. Indeed, for any cartesian theory $\mathbb{T}$, any formula presenting a $\mathbb{T}$-model (equivalently, any $\mathbb{T}$-irreducile formula) is isomorphic, in the syntactic category $\mathcal{C}_{\mathbb{T}}$, to a $\mathbb{T}$-cartesian formula (cf. Remark 6.1.18(b) below).

Corollary 6.1.15 *Let $\mathbb{T}$ be a theory of presheaf type and M a set-based $\mathbb{T}$-model. Then M is finitely presentable if and only if it is finitely presented.*

Proof Every finitely presented model of a geometric theory $\mathbb{T}$ is finitely presentable. Indeed, if M is presented by a formula $\{\vec{x} . \phi\}$ then the hom functor $\mathrm{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(M, -)$: $\mathbb{T}$-mod($\mathbf{Set}$) $\to$ $\mathbf{Set}$ is isomorphic to the functor $[[\vec{x} . \phi]]_{-}$, which preserves filtered colimits since the formula $\phi(\vec{x})$ is geometric (recall that finite limits as well as arbitrary colimits commute with filtered colimits). The converse direction follows from Theorem 6.1.13. $\square$

Remark 6.1.16 The 'only if' part of the corollary is not true for general geometric theories; for instance, in the coherent theory of fields the model $\mathbb{Q}$ is finitely presentable but not finitely presented (cf. section 9.4 below).

6.1.5 *A syntactic description of the finitely presentable models*

For a theory of presheaf type $\mathbb{T}$, it is possible to give an explicit syntactic description of the finitely presentable $\mathbb{T}$-models; specifically, we have the following result:

Theorem 6.1.17 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ and $\{\vec{x}\,.\,\phi\}$ a formula over Σ presenting a $\mathbb{T}$-model. Then this model is isomorphic to the Σ-structure $M_{\{\vec{x}\,.\,\phi\}}$ defined as follows:*

(i) *For any sort A of Σ, $M_{\{\vec{x}\,.\,\phi\}}A$ is equal to the set $\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^A\,.\,\top\})$ of $\mathbb{T}$-provable equivalence classes of $\mathbb{T}$-provably functional geometric formulae from $\{\vec{x}\,.\,\phi\}$ to $\{x^A\,.\,\top\}$.*

(ii) *For any function symbol $f : A_1 \cdots A_n \to B$ of Σ, the function*

$$M_{\{\vec{x}\,.\,\phi\}}f : \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^{A_1}\,.\,\top\}) \times \cdots \times \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^{A_n}\,.\,\top\}) \cong \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^{A_1}, \ldots, x^{A_n}\,.\,\top\}) \to \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^B\,.\,\top\})$$

is equal to $[f] \circ -$ (where $[f] : \{x^{A_1}, \ldots, x^{A_n}\,.\,\top\} \to \{x^B\,.\,\top\}$ is the morphism in $\mathcal{C}_{\mathbb{T}}$ corresponding to f).

(iii) *For any relation symbol $R \rightarrowtail A_1 \cdots A_n$ of Σ, $M_{\{\vec{x}\,.\,\phi\}}R$ is the subobject of*

$$\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^{A_1}\,.\,\top\}) \times \cdots \times \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^{A_n}\,.\,\top\}) \cong \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{x^{A_1}, \ldots, x^{A_n}\,.\,\top\})$$

given by $[R] \circ -$, (where $[R] : \{x^{A_1}, \ldots, x^{A_n}\,.\,R\} \rightarrowtail \{x^{A_1}, \ldots, x^{A_n}\,.\,\top\}$ is the subobject in $\mathcal{C}_{\mathbb{T}}$ corresponding to R).

Proof Recall that we have a canonical equivalence of categories

$$\mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathbf{Set}) \simeq \mathbb{T}\text{-mod}(\mathbf{Set})$$

sending any flat $J_{\mathbb{T}}$-continuous functor $F : \mathcal{C}_{\mathbb{T}} \to \mathbf{Set}$ to the $\mathbb{T}$-model $F(M_{\mathbb{T}})$, where $M_{\mathbb{T}}$ is the universal model of $\mathbb{T}$ in $\mathcal{C}_{\mathbb{T}}$ as in Definition 1.4.4.

We know from (the proof of) Theorem 6.1.13 that for any theory of presheaf type $\mathbb{T}$ over a signature Σ, any formula-in-context $\{\vec{x}\,.\,\phi\}$ over Σ which presents a $\mathbb{T}$-model is $\mathbb{T}$-irreducible, in the sense that every $J_{\mathbb{T}}$-covering sieve on $\{\vec{x}\,.\,\phi\}$ in $\mathcal{C}_{\mathbb{T}}$ is maximal. From this it easily follows that the (flat) representable functor $\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, -) : \mathcal{C}_{\mathbb{T}} \to \mathbf{Set}$ is $J_{\mathbb{T}}$-continuous; indeed, for any $J_{\mathbb{T}}$-covering sieve S on an object $\{\vec{y}\,.\,\psi\}$ of $\mathcal{C}_{\mathbb{T}}$, any arrow $\gamma : \{\vec{x}\,.\,\phi\} \to \{\vec{y}\,.\,\psi\}$ in $\mathcal{C}_{\mathbb{T}}$ factors through one of the arrows belonging to S since the pullback of S along γ is $J_{\mathbb{T}}$-covering and hence maximal. The image $\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, M_{\mathbb{T}})$ under this functor of the universal model $M_{\mathbb{T}}$ of $\mathbb{T}$ in $\mathcal{C}_{\mathbb{T}}$, which clearly coincides with the Σ-structure $M_{\{\vec{x}\,.\,\phi\}}$ in the statement of the theorem, is therefore a $\mathbb{T}$-model. In order to deduce our thesis, it thus remains to verify that the model $M_{\{\vec{x}\,.\,\phi\}}$ satisfies the universal property of the $\mathbb{T}$-model presented by the formula $\{\vec{x}\,.\,\phi\}$, i.e. that for any $\mathbb{T}$-model N in $\mathbf{Set}$, the $\mathbb{T}$-model homomorphisms $M_{\{\vec{x}\,.\,\phi\}} = \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, M_{\mathbb{T}}) \to$

N are in natural bijection with the elements of the set $[[\vec{x} \,.\, \phi]]_N$. But the $\mathbb{T}$-model homomorphisms $\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x} \,.\, \phi\}, M_{\mathbb{T}}) \to N$ are in natural bijection, by the equivalence $\mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathbf{Set}) \simeq \mathbb{T}\text{-mod}(\mathbf{Set})$, with the natural transformations $\mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x} \,.\, \phi\}, -) \to F_N$, that is, by the Yoneda lemma, with the elements of the set $F_N(\{\vec{x} \,.\, \phi\}) = [[\vec{x} \,.\, \phi]]_N$, as required. □

Remarks 6.1.18 (a) If $\mathbb{T}$ is a universal Horn theory and $\vec{x} = (x_1^{A_1}, \ldots, x_n^{A_n})$ then the set $M_{\{\vec{x} \,.\, \phi\}}A$ can be identified with the set of equivalence classes of terms over Σ of type $A_1 \cdots A_n \to A$ modulo the equivalence relation which identifies any two such terms t_1 and t_2 precisely when the sequent $(\phi \vdash_{\vec{x}} t_1 = t_2)$ is provable in $\mathbb{T}$. Indeed, it is shown in [7] (cf. p. 120 therein) that any $\mathbb{T}$-provably functional geometric formula $\theta(\vec{x}, \vec{y})$ between Horn formulae over Σ is $\mathbb{T}$-provably equivalent to a formula of the form $\vec{y} = \vec{t}(\vec{x})$, where $\vec{t}$ is a sequence of terms of the appropriate sorts in the context $\vec{x}$.

(b) If $\mathbb{T}$ is a cartesian theory then any formula presenting a $\mathbb{T}$-model is isomorphic, in the syntactic category $\mathcal{C}_{\mathbb{T}}$, to a $\mathbb{T}$-cartesian formula, and for any such formula $\phi(\vec{x})$, any $\mathbb{T}$-provably functional geometric formula from $\{\vec{x} \,.\, \phi\}$ to $\{x^A \,.\, \top\}$ is $\mathbb{T}$-cartesian, up to $\mathbb{T}$-provable equivalence. This follows immediately from Theorem 6.1.3 by using the explicit description of the universal model of $\mathbb{T}$ in its classifying topos $[(\mathcal{C}_{\mathbb{T}}^{\mathrm{cart}})^{\mathrm{op}}, \mathbf{Set}]$ (cf. Remark 2.1.9).

(c) Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{K}$ a small category of set-based $\mathbb{T}$-models and $\phi(\vec{x})$ a geometric formula over Σ presenting a $\mathbb{T}$-model in $\mathcal{K}$. If the geometric morphism

$$p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

considered in section 5.2.3 has the property that its inverse image $p_{\mathcal{K}}^*$ is full and faithful (for instance, if $p_{\mathcal{K}}$ is hyperconnected) then $\phi(\vec{x})$ is $\mathbb{T}$-irreducible and the argument in the proof of Theorem 6.1.17 applies yielding a syntactic description of the model presented by $\phi(\vec{x})$ as in the statement of Theorem 6.1.17. Indeed, denoting by y and y' the Yoneda embeddings respectively of $\mathcal{C}_{\mathbb{T}}$ into $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ and of $\mathcal{K}^{\mathrm{op}}$ into $[\mathcal{K}, \mathbf{Set}]$, we have that $p_{\mathcal{K}}^*(y\{\vec{x} \,.\, \phi\}) \cong y'(M_{\{\vec{x} \,.\, \phi\}})$. Now, as $y'(M_{\{\vec{x} \,.\, \phi\}})$ is an irreducible object of the topos $[\mathcal{K}, \mathbf{Set}]$ and the property of an object of a topos to be irreducible is reflected by full and faithful inverse images of geometric morphisms, $y\{\vec{x} \,.\, \phi\}$ is an irreducible object of the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$, equivalently $\phi(\vec{x})$ is $\mathbb{T}$-irreducible, as required.

Corollary 6.1.19 *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ and $\phi(\vec{x})$ a geometric formula-in-context over Σ presenting a $\mathbb{T}$-model $M_{\{\vec{x} \,.\, \phi\}}$. Then, for any geometric formula $\psi(\vec{x})$ in the same context, $M_{\{\vec{x} \,.\, \phi\}} \vDash \psi(\vec{\xi_\phi})$ if and only if the sequent $(\phi \vdash_{\vec{x}} \psi)$ is provable in $\mathbb{T}$.*

Proof Let us use the syntactic description of $M_{\{\vec{x} \,.\, \phi\}}$ provided by Theorem 6.1.17. The generators $\vec{\xi_\phi} \in M_{\{\vec{x} \,.\, \phi\}}A_1 \times \cdots \times M_{\{\vec{x} \,.\, \phi\}}A_n$ are given by the canonical projections $\pi_i : \{\vec{x} \,.\, \phi\} \to \{x_i^{A_i} \,.\, \top\}$ (for $i = 1, \ldots, n$). We thus have that

$M_{\{\vec{x}\,.\,\phi\}} \models \psi(\vec{\xi_\phi})$ if and only if $\vec{\xi_\phi} \in [[\vec{x}\,.\,\psi]]_{M_{\{\vec{x}\,.\,\phi\}}} = \mathrm{Hom}_{\mathcal{C}_{\mathbb{T}}}(\{\vec{x}\,.\,\phi\}, \{\vec{x}\,.\,\psi\})$, that is if and only if $< \pi_1, \ldots, \pi_n >: \{\vec{x}\,.\,\phi\} \to \{\vec{x}\,.\,\top\}$ factors through the subobject $\{\vec{x}\,.\,\psi\} \rightarrowtail \{\vec{x}\,.\,\top\}$ in $\mathcal{C}_{\mathbb{T}}$; but this occurs if and only if the sequent $(\phi \vdash_{\vec{x}} \psi)$ is provable in $\mathbb{T}$. □

6.2 Internal finite presentability

In this section we introduce two notions of internal finite presentability for models of geometric theories in Grothendieck toposes, and show that the models of the form $\gamma_{\mathcal{E}}^*(c)$ for a finitely presentable model c of a theory of presheaf type $\mathbb{T}$ satisfy both of them.

6.2.1 *Objects of homomorphisms*

The following result asserts that, for any geometric theory $\mathbb{T}$ and any Grothendieck topos $\mathcal{E}$, the $\mathcal{E}$-indexed category $\underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ introduced in section 5.2.3 is locally small.

Theorem 6.2.1 *Let $\mathbb{T}$ be a geometric theory. Then for any $\mathbb{T}$-models M and N in a Grothendieck topos $\mathcal{E}$ there exists an object $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)$ of $\mathcal{E}$, called the 'object of $\mathbb{T}$-model homomorphisms from M to N', satisfying the universal property that for any object E of $\mathcal{E}$ the generalized elements $E \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)$ are in bijective correspondence, naturally in $E \in \mathcal{E}$, with the $\mathbb{T}$-model homomorphisms $!_E^*(M) \to !_E^*(N)$ in $\mathbb{T}\text{-mod}(\mathcal{E}/E)$.*

Proof Recall from section 5.1.1 that for any small category $\mathcal{C}$ and any Grothendieck topos $\mathcal{E}$, for any two functors $F, G : \mathcal{C} \to \mathcal{E}$ there exists an object $\mathrm{Hom}^{\mathcal{E}}_{\underline{[\mathcal{C},\mathcal{E}]}}(F, G)$ satisfying the property that for any object E of $\mathcal{E}$ the generalized elements $E \to \mathrm{Hom}^{\mathcal{E}}_{\underline{[\mathcal{C},\mathcal{E}]}}(F, G)$ are in bijective correspondence, naturally in $E \in \mathcal{E}$, with the arrows $!_E^* \circ F \to !_E^* \circ G$ in $[\mathcal{C}, \mathcal{E}/E]$, that is, with the natural transformations $!_E^* \circ F \to !_E^* \circ G$. This implies that for any Grothendieck topos $\mathcal{E}$, the indexed category $\underline{[\mathcal{C}, \mathcal{E}]}_{\mathcal{E}}$ of functors on $\mathcal{C}$ with values in $\mathcal{E}$ and natural transformations between them is locally small. It follows in particular that any $\mathcal{E}$-indexed full subcategory of $\underline{[\mathcal{C}, \mathcal{E}]}_{\mathcal{E}}$, such as the indexed category of J-continuous flat functors on $\mathcal{C}$ with values in $\mathcal{E}$, is also locally small.

Now, every geometric theory $\mathbb{T}$ is Morita-equivalent to the theory of flat $J_{\mathbb{T}}$-continuous functors $\mathcal{C}_{\mathbb{T}}$ (cf. section 2.1.2); so the $\mathcal{E}$-indexed category $\underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ is equivalent to the $\mathcal{E}$-indexed category of flat $J_{\mathbb{T}}$-continuous functors on $\mathcal{C}_{\mathbb{T}}$ with values in $\mathcal{E}$. Hence $\underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ is locally small, as required. □

Remarks 6.2.2 Let $\mathbb{T}$ be a geometric theory over a signature Σ. Then

(a) The assignment $(M, N) \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)$ is functorial both in M and N; that is, any homomorphism $f : M \to M'$ in $\mathbb{T}\text{-mod}(\mathcal{E})$ (resp. any homomorphism $g : N' \to N$ in $\mathbb{T}\text{-mod}(\mathcal{E})$) induces an arrow

$$\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(f, N) : \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M', N) \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)$$

(resp. an arrow

$$\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, g) : \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N') \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N))$$

functorially in f (resp. functorially in g).

(b) For any Grothendieck topos $\mathcal{E}$ and $\mathbb{T}$-models M and N in $\mathcal{E}$, we have a canonical embedding

$$\mathrm{Hom}^{\mathcal{F}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N) \to \prod_{A \text{ sort of } \Sigma} NA^{MA}$$

induced by arrows

$$\pi_A : \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N) \to NA^{MA}$$

(for any sort A of Σ) defined in terms of generalized elements as follows: π_A sends any arrow $E \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)$ in $\mathcal{E}$, corresponding to a $\mathbb{T}$-model homomorphism $r :!_E^*(M) \to !_E^*(N)$ in $\mathcal{E}/E$, to the arrow $E \to NA^{MA}$ whose transpose is the arrow $r_A :!_E^*(M)A \to !_E^*(N)A$.

(c) For any $\mathbb{T}$-models M and N in a Grothendieck topos $\mathcal{E}$ and any geometric morphism $f : \mathcal{F} \to \mathcal{E}$, there is a canonical arrow

$$f^*(\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)) \to \mathrm{Hom}^{\mathcal{F}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{F})}}(f^*(M), f^*(N)).$$

Indeed, this arrow corresponds, by the universal property of the object

$$\mathrm{Hom}^{\mathcal{F}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{F})}}(f^*(M), f^*(N)),$$

to the $\mathbb{T}$-model homomorphism

$$f^*(\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)) \times f^*(M) \to f^*(\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)) \times f^*(N)$$

in the topos $\mathcal{F}/f^*(\mathrm{Hom}^{\mathcal{F}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N))$ whose first component, at any sort A of Σ, is the canonical projection and whose second component at A is the arrow

$$f^*(\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N)) \times f^*(MA) \cong f^*(\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N) \times MA) \to f^*(NA)$$

obtained by taking the image under f^* of the arrow

$$\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N) \times MA \to NA$$

given by the transpose of the arrow $\pi_A : \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N) \to NA^{MA}$ defined above.

(d) Since the $\mathcal{E}$-indexed category $\underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ is locally small (cf. Theorem 6.2.1), we have an $\mathcal{E}$-indexed hom functor

$$\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(-,-) : \underline{\mathbb{T}\text{-mod}(\mathcal{E})} \times \underline{\mathbb{T}\text{-mod}(\mathcal{E})} \to \underline{\mathcal{E}}_{\mathcal{E}}.$$

The following proposition gives an explicit description of the generalized elements of the objects of homomorphisms of Theorem 6.2.1 in a particular case of interest. In the statement of the proposition, for a $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, we denote by $\mathrm{Hom}_{\mathcal{E}}(E, M)$ the Σ-structure in **Set** given by the image of M under the product-preserving functor $\mathrm{Hom}_{\mathcal{E}}(E, -) : \mathcal{E} \to \mathbf{Set}$.

Proposition 6.2.3 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, M a model of $\mathbb{T}$ in a Grothendieck topos $\mathcal{E}$, c a set-based $\mathbb{T}$-model and E an object of $\mathcal{E}$. Then the generalized elements $x : E \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$ correspond bijectively to the Σ-structure homomorphisms $\xi_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$.*

Proof By definition of $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$, a generalized element

$$E \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$$

corresponds precisely to a $\mathbb{T}$-model homomorphism $\gamma^*_{\mathcal{E}/E}(c) \to !^*_E(M)$ in the topos $\mathcal{E}/E$. Concretely, such a $\mathbb{T}$-model homomorphism consists of a family of arrows $\tau_A : \gamma^*_{\mathcal{E}/E}(cA) \to !^*_E(MA)$ in $\mathcal{E}/E$ (indexed by the sorts A of Σ) satisfying the conditions defining the notion of Σ-structure homomorphism. Now, each of the arrows $\tau_A : \gamma^*_{\mathcal{E}/E}(cA) \to !^*_E(MA)$ corresponds, via the adjunction between $\gamma^*_{\mathcal{E}/E}$ and the global section functor on the topos $\mathcal{E}/E$, to a function $\xi_A : cA \to \mathrm{Hom}_{\mathcal{E}}(E, MA)$, and it is immediate to see that the above-mentioned conditions translate precisely into the requirement that the functions $\xi_A : cA \to \mathrm{Hom}_{\mathcal{E}}(E, MA)$ should yield a Σ-structure homomorphism $c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$. $\square$

Remark 6.2.4 If the model c is finitely presentable also as a $\mathbb{T}_c$-model, where $\mathbb{T}_c$ is the cartesianization of $\mathbb{T}$ (notice that this is always the case if Σ is finite, c is finite and $\mathbb{T}$ has only a finite number of axioms, cf. Proposition 9.1.1 below) then, for any ($\mathbb{T}_c$-cartesian) formula $\phi(\vec{x})$ over Σ which presents c as a $\mathbb{T}_c$-model, the Σ-structure homomorphisms $c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ are in natural bijection with the elements of the interpretation of the formula $\phi(\vec{x})$ in the Σ-structure $\mathrm{Hom}_{\mathcal{E}}(E, M)$ (cf. Remark 6.2.6(c) below).

6.2.2 *Strong finite presentability*

We shall show in this section that the finitely presentable models of a theory $\mathbb{T}$ of presheaf type enjoy a strong form of syntactic finite presentability with respect to the models of $\mathbb{T}$ in arbitrary Grothendieck toposes.

Let $\mathbb{T}$ be a geometric theory over a signature Σ, c a set-based model of $\mathbb{T}$, $\phi(\vec{x})$ a geometric formula over Σ and $\vec{u}$ an element of $[[\vec{x}\,.\,\phi]]_c$. Then, for any model M of $\mathbb{T}$ in a Grothendieck topos $\mathcal{E}$, there is a canonical arrow

$$\epsilon^M_{(\phi(\vec{x}),\vec{u})} : \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M) \to [[\vec{x}\,.\,\phi]]_M$$

in $\mathcal{E}$, defined in terms of generalized elements as follows: $\epsilon^M_{(\phi(\vec{x}),\vec{u})}$ sends a generalized element

$$E \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M),$$

corresponding under the bijection of Proposition 6.2.3 to a Σ-structure homomorphism $f : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, to the image of $\vec{u}$ under f; notice that this element indeed belongs to $\mathrm{Hom}_{\mathcal{E}}(E, [[\vec{x}\,.\,\phi]]_M)$ since, as f is a Σ-structure homomorphism, the image of $[[\vec{x}\,.\,\phi]]_c$ under f is contained in $[[\vec{x}\,.\,\phi]]_{\mathrm{Hom}_{\mathcal{E}}(E,M)}$, which is in turn contained in $\mathrm{Hom}_{\mathcal{E}}(E, [[\vec{x}\,.\,\phi]]_M)$ as the functor $\mathrm{Hom}_{\mathcal{E}}(E, -)$ is cartesian.

Definition 6.2.5 Let $\mathbb{T}$ be a geometric theory over a signature Σ and c a $\mathbb{T}$-model in **Set**. Then c is said to be *strongly finitely presented* if there exist a geometric formula $\phi(\vec{x})$ over Σ and a finite string of elements $\vec{u}$ of $[[\vec{x}\,.\,\phi]]_c$, called the *strong generators* of c, such that for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ the arrow

$$\epsilon^M_{(\phi(\vec{x}),\vec{u})} : \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M) \to [[\vec{x}\,.\,\phi]]_M$$

is an isomorphism, equivalently if for every $E \in \mathcal{E}$ the Σ-structure homomorphisms $f : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ are in natural bijection with the generalized elements $E \to [[\vec{x}\,.\,\phi]]_M$ via the assignment $f \to f(\vec{u})$.

Remarks 6.2.6 (a) If the equivalent condition in Definition 6.2.5 is satisfied by all models M of $\mathbb{T}$ inside Grothendieck toposes for $E = 1_{\mathcal{E}}$ then it is true in general, by the localization principle.

(b) If the condition in Definition 6.2.5 is satisfied then

$$[[\vec{x}\,.\,\phi]]_{\mathrm{Hom}_{\mathcal{E}}(E,M)} \cong \mathrm{Hom}_{\mathcal{E}}(E, [[\vec{x}\,.\,\phi]]_M)$$

for every $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ and any object E of $\mathcal{E}$.

(c) If a model of a geometric theory $\mathbb{T}$ is finitely presentable as a model of its cartesianization then it is strongly finitely presented. Indeed, any structure of the form $\mathrm{Hom}_{\mathcal{E}}(E, M)$, where M is a model of $\mathbb{T}$ in a Grothendieck topos $\mathcal{E}$, is a model of the cartesianization of $\mathbb{T}$ (as it is obtained by applying a global section functor, which is cartesian, to a model of $\mathbb{T}$), and for any cartesian formula $\phi(\vec{x})$, $[[\vec{x}\,.\,\phi]]_{\mathrm{Hom}_{\mathcal{E}}(E,M)} \cong \mathrm{Hom}_{\mathcal{E}}(E, [[\vec{x}\,.\,\phi]]_M)$.

Proposition 6.2.7 *Let $\mathbb{T}$ be a theory of presheaf type classified by the topos $[\mathcal{K}, \mathbf{Set}]$, where $\mathcal{K}$ is a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**)*. Then for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ and any $\mathbb{T}$-model c in $\mathcal{K}$ there is a natural bijective correspondence between the Σ-structure homomorphisms $c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ and* *the elements of the set* $\mathrm{Hom}_{\mathcal{E}}(E, F_M(c))$*, where* F_M *is the flat functor* $\mathcal{K}^{\mathrm{op}} \to \mathcal{E}$ *corresponding to the model* M *via the canonical Morita equivalence for* $\mathbb{T}$*.*

Proof We can clearly suppose without loss of generality that $E = 1_{\mathcal{E}}$. The adjunction between $\gamma_{\mathcal{E}}^*$ and the global sections functor $\Gamma_{\mathcal{E}} : \mathcal{E} \to \mathbf{Set}$ provides a natural bijective correspondence between the elements of the set $\mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, F_M(c))$ and the natural transformations $\gamma_{\mathcal{E}}^* \circ y_{\mathcal{K}}(c) \to F_M$, where $y_{\mathcal{K}} : \mathcal{K} \to [\mathcal{K}^{\mathrm{op}}, \mathbf{Set}]$ is the Yoneda embedding. Via the canonical Morita equivalence

$$\tau_{\mathcal{E}} : \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

for $\mathbb{T}$, such natural transformations are in natural bijective correspondence with the $\mathbb{T}$-model homomorphisms $\gamma_{\mathcal{E}}^*(c) \cong \tau_{\mathcal{E}}(\gamma_{\mathcal{E}}^* \circ y_{\mathcal{K}}(c)) \to \tau_{\mathcal{E}}(F_M) = M$. But these homomorphisms are, by Proposition 6.2.3, in natural bijective correspondence with the Σ-structure homomorphisms $c \to \mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, M)$, as required. □

Corollary 6.2.8 *Let* $\mathbb{T}$ *be a theory of presheaf type over a signature* Σ *and* $\phi(\vec{x})$ *a geometric formula over* Σ *presenting a* $\mathbb{T}$*-model* $M_{\{\vec{x}\,.\,\phi\}}$*. Then* $M_{\{\vec{x}\,.\,\phi\}}$ *is strongly finitely presented by* $\phi(\vec{x})$*; that is, for any* $\mathbb{T}$*-model* M *in a Grothendieck topos* $\mathcal{E}$ *and any object* E *of* $\mathcal{E}$*, the* Σ*-structure homomorphisms* $M_{\{\vec{x}\,.\,\phi\}} \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ *correspond bijectively to the elements of the set* $\mathrm{Hom}_{\mathcal{E}}(E, [[\vec{x}\,.\,\phi]]_M)$ *via the assignment sending a* Σ*-structure homomorphism* $M_{\{\vec{x}\,.\,\phi\}} \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ *to the image of the strong generators of* $M_{\{\vec{x}\,.\,\phi\}}$ *under it.*

Proof Clearly, we can suppose without loss of generality $E = 1_{\mathcal{E}}$ in the condition of the corollary.

The canonical Morita equivalence

$$\xi^{\mathcal{E}} : \mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

for $\mathbb{T}$ (in the sense of section 6.1.1) sends any $\mathbb{T}$-model M to the flat functor $F_M := \mathrm{Hom}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma_{\mathcal{E}}^*(-), M)$ (cf. the proof of Theorem 6.3.1) and any flat functor $F : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}} \to \mathcal{E}$ to the model $\tilde{F}(M_{\mathbb{T}})$, where $\tilde{F} : \mathcal{C}_{\mathbb{T}} \to \mathcal{E}$ is the extension of F to the syntactic category $\mathcal{C}_{\mathbb{T}}$ (in the sense of section 5.2.3) and $M_{\mathbb{T}}$ is the universal model of $\mathbb{T}$ in $\mathcal{C}_{\mathbb{T}}$ (as in Definition 1.4.4). Indeed, $\xi^{\mathcal{E}}$ is induced by the canonical geometric morphism (in fact, equivalence)

$$p_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})} : [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}),$$

i.e. it is given by the composite of the induced equivalence

$$\mathbf{Flat}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, \mathcal{E}) \simeq \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E})$$

with the canonical equivalence $\mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$ sending any $J_{\mathbb{T}}$-continuous flat functor G on $\mathcal{C}_{\mathbb{T}}$ to the $\mathbb{T}$-model $G(M_{\mathbb{T}})$.

By Theorem 5.2.5, for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, there is an isomorphism $z_{(M, \{\vec{x}\,.\,\phi\})} : F_M(M_{\{\vec{x}\,.\,\phi\}}) \cong \widetilde{F_M}(\{\vec{x}\,.\,\phi\}) = [[\vec{x}\,.\,\phi]]_M$. Thus, applying Proposition 6.2.7 to the model $c = M_{\{\vec{x}\,.\,\phi\}}$, we obtain a bijective correspondence between the Σ-structure homomorphisms $M_{\{\vec{x}\,.\,\phi\}} \to \mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, M)$ and the elements of the set $\mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, [[\vec{x}\,.\,\phi]]_M)$. It remains to show that this correspondence can be identified with the assignment sending a Σ-structure homomorphism $M_{\{\vec{x}\,.\,\phi\}} \to \mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, M)$ to the image of the strong generators of $M_{\{\vec{x}\,.\,\phi\}}$ under it. Recall that the bijection of Proposition 6.2.7 can be described as follows: any Σ-structure homomorphism $f : M_{\{\vec{x}\,.\,\phi\}} \to \mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, M)$ corresponding to a $\mathbb{T}$-model homomorphism $\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}}) \to M$ and hence, via $\xi^{\mathcal{E}}$, to a natural transformation $\gamma_{\mathcal{E}}^* \circ y(M_{\{\vec{x}\,.\,\phi\}}) \cong F_{\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}})} \to F_M$ (where y is the Yoneda embedding $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\mathrm{op}} \hookrightarrow [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$), is sent to the global element of $F_M(M_{\{\vec{x}\,.\,\phi\}})$ corresponding via the Yoneda lemma to this transformation. To deduce our thesis, it thus remains to verify that the canonical isomorphism of functors $\gamma_{\mathcal{E}}^* \circ y(M_{\{\vec{x}\,.\,\phi\}}) = \gamma_{\mathcal{E}}^* \circ F_{M_{\{\vec{x}\,.\,\phi\}}} \cong F_{\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}})}$, when evaluated at $M_{\{\vec{x}\,.\,\phi\}}$ and composed with the isomorphism $z_{(\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}}),\{\vec{x}\,.\,\phi\})} : F_{\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}})}(M_{\{\vec{x}\,.\,\phi\}}) \cong [[\vec{x}\,.\,\phi]]_{\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}})} \cong \gamma_{\mathcal{E}}^*([[\vec{x}\,.\,\phi]]_{M_{\{\vec{x}\,.\,\phi\}}})$, sends the coproduct component of $\gamma_{\mathcal{E}}^*((y(M_{\{\vec{x}\,.\,\phi\}}))(M_{\{\vec{x}\,.\,\phi\}}))$ corresponding to the identity on $M_{\{\vec{x}\,.\,\phi\}}$ to the coproduct component of $\gamma_{\mathcal{E}}^*([[\vec{x}\,.\,\phi]]_{M_{\{\vec{x}\,.\,\phi\}}})$ corresponding to the strong generators of $M_{\{\vec{x}\,.\,\phi\}}$. By the naturality in $\mathcal{E}$ of $\xi^{\mathcal{E}}$, we can clearly suppose, without loss of generality, that $\mathcal{E}$ is equal to **Set**. But from the proof of Theorem 6.1.17 it is clear that the strong generators of $M_{\{\vec{x}\,.\,\phi\}}$ correspond to the identity on $\{\vec{x}\,.\,\phi\}$ via the Yoneda lemma and hence to the identity on $M_{\{\vec{x}\,.\,\phi\}}$ via the above-mentioned bijection, as required. □

Remark 6.2.9 We can express the bijective correspondence of Corollary 6.2.8 by saying that for any Grothendieck topos $\mathcal{E}$ the $\mathcal{E}$-indexed functor $[[\vec{x}\,.\,\phi]]_{-} : \underline{\mathbb{T}\text{-mod}(\mathcal{E})} \to \underline{\mathcal{E}}_{\mathcal{E}}$ assigning to any $\mathbb{T}$-model M the interpretation of $\phi(\vec{x})$ in M is represented as an $\mathcal{E}$-indexed functor by the object $\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}})$, in the sense of being isomorphic to the $\mathcal{E}$-representable functor $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma_{\mathcal{E}}^*(M_{\{\vec{x}\,.\,\phi\}}), -)$.

Proposition 6.2.10 *Let $\mathbb{T}$ be a theory of presheaf type and $\mathbb{S}$ be a subtheory of $\mathbb{T}$ (that is, a theory $\mathbb{S}$ of which $\mathbb{T}$ is a quotient) such that every set-based model of $\mathbb{S}$ admits a representation as the structure* $\mathrm{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, M)$ *of global sections of a $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$. Then every finitely presentable model of $\mathbb{T}$ is finitely presented as a model of $\mathbb{S}$.*

Proof If a theory $\mathbb{T}$ is of presheaf type then any finitely presentable model of $\mathbb{T}$ is strongly finitely presented (by Corollary 6.2.8) and hence, by Remark 6.2.6(b), finitely presented relatively to whatever subtheory $\mathbb{S}$ of $\mathbb{T}$ whose set-based models admit representations as sections $\mathrm{Hom}_{\mathcal{E}}(E, M)$ of models M of $\mathbb{T}$ in Grothendieck toposes. □

Remarks 6.2.11 (a) Pairs of theories satisfying the hypotheses of the proposition are investigated for instance in [34] and [38].

(b) The proposition can often be profitably applied to the cartesianization of $\mathbb{T}$; however, it is not known if it is always the case that every model of it admits a representation of the above kind (cf. also Remark 8.2.2(c)).

6.2.3 *Semantic $\mathcal{E}$-finite presentability*

In this section, we introduce a semantic notion of $\mathcal{E}$-finite presentability of a model of a geometric theory in a Grothendieck topos $\mathcal{E}$, which generalizes the classical notion in the theory of finitely accessible categories, and show that all the 'constant' finitely presentable models of a theory of presheaf type in a Grothendieck topos $\mathcal{E}$ are $\mathcal{E}$-finitely presentable.

Recall from section 5.2.3 that, for any geometric theory $\mathbb{T}$ over a signature Σ and any Grothendieck topos $\mathcal{E}$, the $\mathcal{E}$-indexed category $\underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ admits $\mathcal{E}$-indexed colimits of diagrams defined on $\mathcal{E}$-final, $\mathcal{E}$-filtered subcategories of a small $\mathcal{E}$-indexed category, which are jointly created by the forgetful functors $U_A : \underline{\mathbb{T}\text{-mod}(\mathcal{E})} \to \underline{\mathcal{E}}_{\mathcal{E}}$ corresponding to the sorts A of Σ. In particular, an $\mathcal{E}$-indexed cocone (M, μ) over a diagram $D : \underline{\mathcal{A}}_{\mathcal{E}} \to \underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ defined on a $\mathcal{E}$-final, $\mathcal{E}$-filtered subcategory of a small $\mathcal{E}$-indexed category is colimiting if and only if, for every sort A of Σ, $U_A((M, \mu))$ is a colimiting cocone over the diagram $U_A \circ D$ in $\underline{\mathcal{E}}_{\mathcal{E}}$.

Definition 6.2.12 Let $\mathbb{T}$ be a geometric theory and M be a model of $\mathbb{T}$ in a Grothendieck topos $\mathcal{E}$. Then M is said to be *$\mathcal{E}$-finitely presentable* if the $\mathcal{E}$-indexed functor $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, -) : \underline{\mathbb{T}\text{-mod}(\mathcal{E})} \to \underline{\mathcal{E}}_{\mathcal{E}}$ of section 6.2.1 preserves $\mathcal{E}$-filtered colimits (of diagrams defined on $\mathcal{E}$-final, $\mathcal{E}$-filtered subcategories of a small $\mathcal{E}$-indexed category).

Proposition 6.2.13 *Let $\mathbb{T}$ be a theory of presheaf type and c a finitely presentable $\mathbb{T}$-model. Then, for any Grothendieck topos $\mathcal{E}$, the $\mathbb{T}$-model $\gamma^*_{\mathcal{E}}(c)$ is $\mathcal{E}$-finitely presentable.*

Proof If c is presented by a geometric formula $\phi(\vec{x})$ over the signature Σ of $\mathbb{T}$, by Corollary 6.2.8 the functor $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$ is naturally isomorphic to the functor $[[\vec{x} \,.\, \phi]]_{-} : \underline{\mathbb{T}\text{-mod}(\mathcal{E})} \to \underline{\mathcal{E}}_{\mathcal{E}}$, which, as we observed in section 5.2.3, preserves $\mathcal{E}$-filtered colimits of diagrams defined on $\mathcal{E}$-final and $\mathcal{E}$-filtered subcategories of a small $\mathcal{E}$-indexed category. □

Remark 6.2.14 The proof of the proposition shows that, more generally, for any strongly finitely presentable model c of a geometric theory $\mathbb{T}$ in the sense of Definition 6.2.5 (for instance, a finite model of $\mathbb{T}$ if the signature of $\mathbb{T}$ is finite – note that such a model is finitely presented with respect to the empty theory over the signature of $\mathbb{T}$ by Proposition 9.1.1, whence Remark 6.2.6(c) applies) and any Grothendieck topos $\mathcal{E}$, the $\mathbb{T}$-model $\gamma^*_{\mathcal{E}}(c)$ is $\mathcal{E}$-finitely presentable.

The following proposition provides an explicit characterization of the $\mathcal{E}$-finitely presentable models of a geometric theory $\mathbb{T}$.

Proposition 6.2.15 *Let $\mathbb{T}$ be a geometric theory, $\mathcal{E}$ a Grothendieck topos and c a $\mathbb{T}$-model in $\mathcal{E}$. Then c is $\mathcal{E}$-finitely presentable if and only if for every $\mathcal{E}$-indexed diagram $D : \underline{\mathcal{A}}_{\mathcal{E}} \to \underline{\mathbb{T}\text{-mod}(\mathcal{E})}$ defined on an $\mathcal{E}$-filtered, $\mathcal{E}$-strictly final subcategory $\underline{\mathcal{A}}_{\mathcal{E}}$ of a small $\mathcal{E}$-indexed category with $\mathcal{E}$-indexed colimiting cocone (M, μ) (and colimit arrows $\mu_{(E,x)} : D_E(x) \to !^*_E(M)$), the following conditions are verified:*

(i) *For any object E of $\mathcal{E}$ and $\mathbb{T}$-model homomorphism $h :!_E^*(c) \to !_E^*(M)$ in the topos $\mathcal{E}/E$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ an object x_i of $\mathcal{A}_{E_i}$ and a $\mathbb{T}$-model homomorphism $\alpha_i :!_{E_i}^*(c) \to D_{E_i}(x_i)$ in the topos $\mathcal{E}/E_i$ such that $\mu_{(E_i,x_i)} \circ \alpha_i = e_i^*(h)$.*

(ii) *For any pairs (x, y) and (x', y'), where x and x' are objects of $\mathcal{A}_E$, y is a $\mathbb{T}$-model homomorphism $!_F^*(c) \to f^*(D_E(x))$ in $\mathcal{E}/F$ and y' is a $\mathbb{T}$-model homomorphism $!_F^*(c) \to f^*(D_E(x'))$ in $\mathcal{E}/F$ (for an arrow $f : F \to E$ in $\mathcal{E}$), we have $f^*(\mu_E(x)) \circ y = f^*(\mu_E(x')) \circ y'$ if and only if there exist an epimorphic family $\{f_i : F_i \to F \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ arrows $g_i : \mathcal{A}_{f \circ f_i}(x) \to z_i$ and $h_i : \mathcal{A}_{f \circ f_i}(x') \to z_i$ in the category $\mathcal{A}_{F_i}$ such that $D_{F_i}(g_i) \circ f_i^*(y) = D_{F_i}(h_i) \circ f_i^*(y')$.*

Proof The proposition follows as an immediate consequence of Corollary 5.1.23, applied to the composite $\mathcal{E}$-indexed functor $\underline{\mathrm{Hom}^{\mathcal{E}}_{\mathbb{T}\text{-mod}(\mathcal{E})}}(c, -) \circ D$. □

6.3 Semantic criteria for a theory to be of presheaf type

In this section we establish our main characterization theorem providing necessary and sufficient conditions for a geometric theory to be of presheaf type. These conditions are entirely expressed in terms of the models of the theory in arbitrary Grothendieck toposes and are fully constructive.

We shall first prove the theorem and then proceed to reformulate its conditions in more concrete terms so as to make them easier to check in practice.

6.3.1 *The characterization theorem*

Recall from section 6.1.1 that the classifying topos of a theory of presheaf type $\mathbb{T}$ can be canonically represented as the topos [f.p.$\mathbb{T}$-mod(**Set**), **Set**] of set-valued functors on the category f.p.$\mathbb{T}$-mod(**Set**) of finitely presentable $\mathbb{T}$-models in **Set**; this is not the only possible representation of the classifying topos of $\mathbb{T}$ as a presheaf topos, but for any small category $\mathcal{K}$, $\mathbb{T}$ is classified by the topos $[\mathcal{K}, \mathbf{Set}]$ if and only if the Cauchy completion of $\mathcal{K}$ is equivalent to f.p.$\mathbb{T}$-mod(**Set**).

Theorem 6.3.1 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**). *Then $\mathbb{T}$ is a theory of presheaf type classified by the topos $[\mathcal{K}, \mathbf{Set}]$ if and only if all of the following conditions are satisfied:*

(i) *For any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, the functor*

$$H_M := \underline{\mathrm{Hom}^{\mathcal{E}}_{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma_{\mathcal{E}}^*(-), M) : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$$

is flat.

(ii) *For any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, if the functor $H_M : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$ in condition (i) is flat then its extension $\widetilde{H_M} : \mathcal{C}_{\mathbb{T}} \to \mathcal{E}$ to the syntactic category $\mathcal{C}_{\mathbb{T}}$ (in the sense of section 5.2.3) satisfies the property that the canonical morphism $\varepsilon^M : \widetilde{H_M}(M_{\mathbb{T}}) \to M$ (given by the counit of the adjunction of Theorem 5.2.12 via the canonical equivalence* $\mathbb{T}$-mod$(\mathcal{E}) \simeq$ $\mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E})$ *– cf. Remark 5.2.13(b)) is an isomorphism.*

(iii) *Any of the following (equivalent, if (i) and (ii) hold) conditions is satisfied:*

(a) *For any model c in $\mathcal{K}$, $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ and geometric morphism $f : \mathcal{F} \to \mathcal{E}$, the canonical morphism*

$$f^*(\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)) \to \mathrm{Hom}^{\mathcal{F}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{F})}}(\gamma^*_{\mathcal{F}}(c), f^*(M)),$$

provided by Remark 6.2.2(c) via the identification $\gamma^*_{\mathcal{F}}(c) \cong f^*(\gamma^*_{\mathcal{E}}(c))$, is an isomorphism.

(b) *For any flat functor $F : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$, the canonical natural transformation*

$$\eta^F : F \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(-), \tilde{F}(M_{\mathbb{T}})) \cong \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}},\mathcal{E})}}(\gamma^*_{\mathcal{E}} \circ \widetilde{y_{\mathcal{K}}(-)}, \tilde{F})$$

of Theorem 5.2.12 is an isomorphism, where $y_{\mathcal{K}} : \mathcal{K} \to \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathbf{Set})$ is the Yoneda embedding.

(c) *The functor*

$$u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})} : \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}_{J_{\mathbb{T}}}(\mathcal{C}_{\mathbb{T}}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

of section 5.2.3 is full and faithful.

Proof Let us begin by proving that conditions (i), (ii) and (iii) are all necessary for $\mathbb{T}$ to be classified by the topos $[\mathcal{K}, \mathbf{Set}]$.

As remarked in section 6.1.1, if $\mathbb{T}$ is of presheaf type classified by the topos $[\mathcal{K}, \mathbf{Set}]$ then we have a Morita equivalence

$$\tau_{\mathcal{E}} : \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

which can be supposed to be canonical without loss of generality, i.e. to send any finitely presentable $\mathbb{T}$-model c in $\mathcal{K}$ to the functor $\gamma^*_{\mathcal{E}} \circ y_{\mathcal{K}}(c)$. It follows that, for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ corresponding to a flat functor F_M under the Morita equivalence $\tau_{\mathcal{E}}$, the object $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$ is isomorphic to the object $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbf{Flat}(\mathcal{K}^{\mathrm{op}},\mathcal{E})}}(\gamma^*_{\mathcal{E}} \circ y_{\mathcal{K}}(c), F_M) \cong F_M(c)$, naturally in M and in c (cf. Corollary 5.3.2). Therefore the functor

$$\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(-), M) : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$$

is flat, it being isomorphic to F_M. This proves that condition (i) of the theorem is satisfied.

We observed in the proof of Corollary 6.2.8 that the functor

$$\mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \to \mathbb{T}\text{-mod}(\mathcal{E})$$

forming the canonical Morita equivalence $\tau_{\mathcal{E}}$ for $\mathbb{T}$ sends any flat functor $F : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$, to the $\mathbb{T}$-model $\tilde{F}(M_{\mathbb{T}})$ (where $M_{\mathbb{T}}$ is the universal model of $\mathbb{T}$ in $\mathcal{C}_{\mathbb{T}}$ as in Definition 1.4.4).

The fact that $\tau_{\mathcal{E}}$ is an equivalence thus implies that the canonical morphism $\widetilde{H_M}(M_{\mathbb{T}}) \to M$ is an isomorphism. This shows that condition (ii) of the theorem is satisfied.

The fact that conditions (iii)(b) and (iii)(c) hold also follows at once from the fact that $\tau_{\mathcal{E}}$ is an equivalence.

We have thus proved that conditions (i), (ii), (iii)(b) and (iii)(c) of the theorem are necessary for $\mathbb{T}$ to be classified by the topos $[\mathcal{K}, \mathbf{Set}]$.

Next, we notice that condition (iii)(b) implies condition (iii)(a) under the assumption that conditions (i) and (ii) hold. Indeed, for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, condition (i) ensures that the functor $H_M : \mathcal{K}^{\text{op}} \to \mathcal{E}$ is flat; for any geometric morphism $f : \mathcal{F} \to \mathcal{E}$, applying condition (iii)(b) to the flat functor $f^* \circ H_M$ and invoking condition (ii) thus yields, in view of the naturality in $\mathcal{E}$ of the operation $\widetilde{(-)}$, a natural isomorphism between the flat functor $f^* \circ H_M$ and the functor $H_{f^*(M)}$.

Since conditions (iii)(b) and (iii)(c) are equivalent by Remarks 5.2.13(b)-(c), to complete the proof of the theorem it remains to show that conditions (i), (ii) and (iii)(a) imply all together that $\mathbb{T}$ is classified by the presheaf topos $[\mathcal{K}, \mathbf{Set}]$.

Under conditions (i), (ii) and (iii)(a), we can define functors

$$G_{\mathcal{E}} : \mathbf{Flat}(\mathcal{K}^{\text{op}}, \mathcal{E}) \to \mathbb{T}\text{-mod}(\mathcal{E})$$

and

$$H_{\mathcal{E}} : \mathbb{T}\text{-mod}(\mathcal{E}) \to \mathbf{Flat}(\mathcal{K}^{\text{op}}, \mathcal{E}),$$

for each Grothendieck topos $\mathcal{E}$, which are natural in $\mathcal{E}$ and inverse to each other up to isomorphism, as follows.

We set $H_{\mathcal{E}}(M)$ equal to the functor

$$H_M = \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(-), M) : \mathcal{K}^{\text{op}} \to \mathcal{E}.$$

This assignment is natural in M (cf. Remark 6.2.2) and hence defines a functor $H_{\mathcal{E}} : \mathbb{T}\text{-mod}(\mathcal{E}) \to \mathbf{Flat}(\mathcal{K}^{\text{op}}, \mathcal{E})$ which is natural in $\mathcal{E}$ by condition (iii)(a).

In the converse direction, for any flat functor $F : \mathcal{K}^{\text{op}} \to \mathcal{E}$, we set $G_{\mathcal{E}}(F) = \tilde{F}(M_{\mathbb{T}})$. Clearly, this assignment is natural in F and hence defines a functor $G_{\mathcal{E}} : \mathbf{Flat}(\mathcal{K}^{\text{op}}, \mathcal{E}) \to \mathbb{T}\text{-mod}(\mathcal{E})$.

The functors $G_{\mathcal{E}}$ and $H_{\mathcal{E}}$ are natural in $\mathcal{E}$ and hence respectively induce geometric morphisms

$$G : [\mathcal{K}, \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

and

$$H : \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \to [\mathcal{K}, \mathbf{Set}].$$

Notice that G coincides with the morphism $p_{\mathcal{K}}$ canonically induced by the universal property of the classifying topos of $\mathbb{T}$ as in section 5.2.3.

We have to prove that

$$H \circ G \cong 1_{[\mathcal{K}, \mathbf{Set}]},$$

or equivalently that $G^* \circ H^* \circ y_{\mathcal{K}} \cong y_{\mathcal{K}}$.

Let us consider the geometric morphisms

$$e_N : \mathbf{Set} \to [\mathcal{K}, \mathbf{Set}]$$

corresponding to the objects N of $\mathcal{K}$. We shall prove that $G^* \circ H^* \circ y_{\mathcal{K}} \cong y_{\mathcal{K}}$ by showing that $e_N^* \circ G^* \circ H^* \circ y_{\mathcal{K}} \cong e_N^* \circ y_{\mathcal{K}} = \mathrm{Hom}_{\mathcal{K}}(-, N)$ naturally in $N \in \mathcal{K}$.

Let us denote by M_G the $\mathbb{T}$-model in $[\mathcal{K}, \mathbf{Set}]$ corresponding to the geometric morphism G. Clearly, for any $\mathbb{T}$-model N in $\mathcal{K}$, $e_N^*(M_G) \cong N$.

For any $\mathbb{T}$-model P in $\mathcal{E}$, $H_{\mathcal{E}}(P) = f_P^* \circ H^* \circ y'_{\mathcal{K}}$, where $f_P : \mathcal{E} \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is the geometric morphism corresponding to P via the universal property of the classifying topos of $\mathbb{T}$ and $y'_{\mathcal{K}} : \mathcal{K}^{\mathrm{op}} \to [\mathcal{K}, \mathbf{Set}]$ is the Yoneda embedding.

We have $H_{[\mathcal{K}, \mathbf{Set}]}(M_G) = G^* \circ H^* \circ y'_{\mathcal{K}}$. But

$$H_{[\mathcal{K}, \mathbf{Set}]}(M_G) = \mathrm{Hom}^{[\mathcal{K}, \mathbf{Set}]}_{\underline{\mathbb{T}\text{-mod}([\mathcal{K}, \mathbf{Set}])}}(\gamma^*_{[\mathcal{K}, \mathbf{Set}]}(-), M_G)$$

and, by condition (iii)(a),

$$\begin{aligned} e_N^*(\mathrm{Hom}^{[\mathcal{K}, \mathbf{Set}]}_{\underline{\mathbb{T}\text{-mod}([\mathcal{K}, \mathbf{Set}])}}(\gamma^*_{[\mathcal{K}, \mathbf{Set}]}(-), M_G)) &\cong \mathrm{Hom}^{\mathbf{Set}}_{\underline{\mathbb{T}\text{-mod}(\mathbf{Set})}}(-, e_N^*(M_G)) \\ &\cong \mathrm{Hom}_{\mathcal{K}}(-, N), \end{aligned}$$

as required. This proves that $H \circ G \cong 1_{[\mathcal{K}, \mathbf{Set}]}$. On the other hand, by condition (ii), $G \circ H$ is isomorphic to the identity. We can thus conclude that $\mathbb{T}$ is classified by the topos $[\mathcal{K}, \mathbf{Set}]$, as required. □

Remarks 6.3.2 (a) Conditions (i) and (ii) of Theorem 6.3.1 together imply that the canonical geometric morphism

$$p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

(cf. section 5.2.3) is a surjection, equivalently that the models in $\mathcal{K}$ are jointly conservative for $\mathbb{T}$. Indeed, by condition (i) the functor H_U, where U is 'the' universal model of $\mathbb{T}$ lying in its classifying topos, is flat. By applying Corollary 5.2.7 to it, we obtain that for any geometric sequent $\sigma \equiv (\phi \vdash_{\vec{x}} \psi)$ over Σ which is valid in every $\mathbb{T}$-model in $\mathcal{K}$, $\widetilde{H_U}(\{\vec{x} \,.\, \phi\}) \leq \widetilde{H_U}(\{\vec{x} \,.\, \psi\})$; condition (ii), combined with the conservativity of U, thus allows us to conclude that σ is provable in $\mathbb{T}$, as required.

(b) The following condition is sufficient, together with conditions (i) and (ii) of the theorem (or equivalently, together with condition (i) and the requirement that the models in $\mathcal{K}$ be jointly conservative for $\mathbb{T}$), for $\mathbb{T}$ to be classified by the topos $[\mathcal{K}, \mathbf{Set}]$, but necessary only if one assumes the axiom of choice:
(∗) There is an assignment $M \to \phi_M(\overrightarrow{x_M})$ sending a $\mathbb{T}$-model M in $\mathcal{K}$ to a geometric formula-in-context $\phi_M(\overrightarrow{x_M})$ presenting it such that every $\mathbb{T}$-model

homomorphism $M \to N$ between models in $\mathcal{K}$ is induced by a $\mathbb{T}$-provably functional formula from $\phi_N(\overrightarrow{x_N})$ to $\phi_M(\overrightarrow{x_M})$.

The necessity of condition $(*)$ (under the axiom of choice) follows from Theorem 6.1.13. To prove its sufficiency, let us show that, under conditions (i) and (ii), or under condition (i) and the assertion that the models in $\mathcal{K}$ are jointly conservative for $\mathbb{T}$, condition $(*)$ implies condition (iii)(c). Under either of these assumptions, the canonical geometric morphism $p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \to \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is a surjection (cf. Remark 6.3.2(a)). From this it follows that, for any geometric formulae $\{\vec{x} \,.\, \phi\}$ and $\{\vec{y} \,.\, \psi\}$ over Σ respectively presenting models c and d in $\mathcal{K}$, there can be at most one $\mathbb{T}$-provably functional formula $\{\vec{x} \,.\, \phi\} \to \{\vec{y} \,.\, \psi\}$ over Σ, up to $\mathbb{T}$-provable equivalence, inducing a given homomorphism of $\mathbb{T}$-models $d \to c$. Condition $(*)$ thus yields a (full and faithful) functor $R : \mathcal{K}^{\mathrm{op}} \to \mathcal{C}_{\mathbb{T}}$ such that $u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}(F) \circ R \cong F$ naturally in F (cf. Theorem 5.2.5 and Remark 5.2.6), making condition (iii)(c) satisfied.

(c) Conditions (ii) and (iii)(c) of Theorem 6.3.1 admit invariant formulations, which can be profitably exploited in the presence of different representations for the classifying topos of $\mathbb{T}$. Indeed, they can both be entirely reformulated in terms of the extension of flat functors operation (in the sense of section 5.2.1) along the canonical geometric morphism

$$p_{\mathcal{K}} : [\mathcal{K}, \mathbf{Set}] \to \mathbf{Set}[\mathbb{T}].$$

Specifically, any (essentially small) site of definition $(\mathcal{C}, J)$ for $\mathbf{Set}[\mathbb{T}]$ gives rise to a functor

$$G^{(\mathcal{C},J)}_{\mathcal{E}} : \mathbf{Flat}(\mathcal{K}^{\mathrm{op}}, \mathcal{E}) \to \mathbf{Flat}_J(\mathcal{C}, \mathcal{E})$$

(note that if $(\mathcal{C}, J) = (\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ then $G^{(\mathcal{C},J)}_{\mathcal{E}}$ is the functor $u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}$ defined in section 5.2.3). Condition (iii)(c) asserts that this functor, whose explicit description is given in section 5.2.1, is full and faithful, while condition (ii) asserts that, denoting by

$$u_{\mathcal{E}} : \mathbf{Flat}_J(\mathcal{C}, \mathcal{E}) \simeq \mathbb{T}\text{-mod}(\mathcal{E})$$

the equivalence canonically induced by the universal property of the classifying topos of $\mathbb{T}$, the canonical morphism $u_{\mathcal{E}}(G^{(\mathcal{C},J)}_{\mathcal{E}}(H_M)) \to M$ is an isomorphism for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$.

(d) Condition (ii) of Theorem 6.3.1 is satisfied by every $\mathbb{T}$-model M in $\mathcal{K}$. Indeed, for any $M \in \mathcal{K}$, the canonical morphism $u_{\mathbf{Set}}(u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}(H_M)) \to M$ of Remark 6.3.2(c) is clearly an isomorphism.

(e) Under condition (i) of Theorem 6.3.1, if condition (iii)(a) is satisfied and the models in $\mathcal{K}$ are jointly conservative for $\mathbb{T}$ then $\mathbb{T}$ satisfies condition (ii). To prove this, we observe that the assertion that the models in $\mathcal{K}$ be jointly conservative for the theory $\mathbb{T}$ is equivalent to the requirement that the geometric morphism $p_{\mathcal{K}}$ be a surjection. By the naturality in $\mathcal{E}$ of the operation $\widetilde{(-)}$ and of the assignment $\mathcal{E} \to H_{\mathcal{E}}$ (notice that the latter follows from condition (iii)(a)), it suffices to verify, since $p_{\mathcal{K}}$ is surjective, that the canonical morphism $\widetilde{H_M}(M_{\mathbb{T}}) \to M$ is an isomorphism for M equal to a model in $\mathcal{K}$; but this is true by Remark 6.3.2(d).

(f) Condition (iii)(b) can be split into two separate conditions: η^F is pointwise monic and η^F is pointwise epic. We shall refer to the first condition as condition (iii)(b)-(1) and to the second as condition (iii)(b)-(2). By Theorem 5.2.12 and Lemma 7.1.9, condition (iii)(b)-(1) is equivalent to the requirement that the functor $u^{\mathbb{T}}_{(\mathcal{K},\mathcal{E})}$ be faithful.

(g) If for every sort A of Σ the formula $\{x^A \,.\, \top\}$ strongly presents a $\mathbb{T}$-model F_A in $\mathcal{K}$ (in the sense of Definition 6.2.5) then condition (i) of the theorem is automatically satisfied; indeed, under this hypothesis, for every sort A of Σ the canonical arrow $\widetilde{H_M}(M_{\mathbb{T}})A \cong \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(F_A), M) \to MA$ is an isomorphism (cf. Theorem 5.2.5).

(h) As observed in the proof of the theorem, the equivalence of conditions (iii)(b) and (iii)(c) holds in general (that is, not only under the assumption that (i) and (ii) hold).

Proposition 6.3.3 *Condition (iii)(c) of Theorem 6.3.1 is always satisfied for* $\mathcal{E} = \mathbf{Set}$.

Proof Let us prove that for any small category $\mathcal{C}$ and any functor $F : \mathcal{C} \to \mathcal{D}$, where $\mathcal{D}$ is a category with filtered colimits, if F is full and faithful and takes values in the full subcategory of $\mathcal{D}$ on the finitely presentable objects then its canonical extension $F' : \text{Ind-}\mathcal{C} = \mathbf{Flat}(\mathcal{C}^{\mathrm{op}}, \mathbf{Set}) \to \mathcal{D}$ is full and faithful. This will imply our thesis by taking F to be the inclusion functor $\mathcal{K} \hookrightarrow \mathbb{T}\text{-mod}(\mathbf{Set})$.

For any $P \in \mathbf{Flat}(\mathcal{C}^{\mathrm{op}}, \mathbf{Set})$, $F'(P) = \mathrm{colim}(F \circ \pi_P)$, where $\pi_P : \int P \to \mathcal{C}$ is the fibration associated with P.

To prove that F' is faithful, we have to show that, for any natural transformations $\alpha, \beta : P \to Q$, $F'(\alpha) = F'(\beta)$ implies $\alpha = \beta$. Since every $P \in \mathbf{Flat}(\mathcal{C}^{\mathrm{op}}, \mathbf{Set})$ can be covered by representables, we can suppose P to be a representable $y_{\mathcal{C}}(c)$ (where $y_{\mathcal{C}} : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ is the Yoneda embedding) without loss of generality. Now, since $F(c)$ is finitely presentable in $\mathcal{D}$, by Lemma 6.1.12 we can suppose without loss of generality that Q is a representable as well. The faithfulness of F' thus follows from that of F.

To prove the fullness of F', we have to show that for any natural transformation $\beta : F'(P) \to F'(Q)$ there exists a natural transformation $\alpha : P \to Q$ such that $F'(\alpha) = \beta$. Thanks to the faithfulness of F', we can clearly suppose P to be a representable $y_{\mathcal{C}}(c)$ without loss of generality. But since $F'(y_{\mathcal{C}}(c)) \cong F(c)$ is finitely presentable in $\mathcal{D}$ and F' preserves filtered colimits, by Lemma 6.1.12 we can suppose without loss of generality Q to be a representable as well. The fullness of F' thus follows from that of F. $\square$

6.3.2 *Concrete reformulations*

In this section we shall give 'concrete' reformulations of the conditions of Theorem 6.3.1 in full generality as well as in some particular cases in which they admit relevant simplifications.

Condition (i)

Theorem 6.3.4 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**)*, $\mathcal{E}$ a Grothendieck topos with a separating family of objects $\mathcal{S}$ and M a $\mathbb{T}$-model in $\mathcal{E}$. Then the following conditions are equivalent:*

(i) *The functor*

$$H_M := \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(-), M) : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$$

is flat.

(ii) (a) *There exist an epimorphic family $\{E_i \to 1_{\mathcal{E}} \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model c_i in $\mathcal{K}$ and a Σ-structure homomorphism $c_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$.*

(b) *For any $\mathbb{T}$-models c and d in $\mathcal{K}$ and Σ-structure homomorphisms $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ and $y : d \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ (where $E \in \mathcal{S}$), there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model b_i in $\mathcal{K}$, $\mathbb{T}$-model homomorphisms $u_i : c \to b_i$, $v_i : d \to b_i$ and a Σ-structure homomorphism $z_i : b_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ x = z_i \circ u_i$ and $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ y = z_i \circ v_i$.*

(c) *For any two homomorphisms $u, v : d \to c$ of $\mathbb{T}$-models in $\mathcal{K}$ and any Σ-structure homomorphism $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ in $\mathcal{E}$ (where $E \in \mathcal{S}$) for which $x \circ u = x \circ v$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a homomorphism $w_i : c \to b_i$ of $\mathbb{T}$-models in $\mathcal{K}$ and a Σ-structure homomorphism $y_i : b_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $w_i \circ u = w_i \circ v$ and $y_i \circ w_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ x$.*

Proof This follows immediately from Proposition 6.2.3 in view of the characterization of flat functors as filtering functors (cf. Proposition 1.1.32). □

Remarks 6.3.5 (a) If for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ and any object E of $\mathcal{E}$ the Σ-structure $\mathrm{Hom}_{\mathcal{E}}(E, M)$ is a $\mathbb{T}$-model then the three conditions (ii)(a), (ii)(b) and (ii)(c) of Theorem 6.3.4 hold if they are satisfied for $\mathcal{E} = \mathbf{Set}$.

(b) Condition (ii)(c) of Theorem 6.3.4 is automatically satisfied if all the $\mathbb{T}$-model homomorphisms in any Grothendieck topos are monic. Indeed, by Proposition 6.2.3, the Σ-structure homomorphisms $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ correspond precisely to the $\mathbb{T}$-model homomorphisms $\overline{x} : \gamma^*_{\mathcal{E}/E}(c) \to !^*_E(M)$. The condition $x \circ u = x \circ v$ is therefore equivalent to $\overline{x} \circ \gamma^*_{\mathcal{E}/E}(u) = \overline{x} \circ \gamma^*_{\mathcal{E}/E}(v)$ and hence, if $\overline{x}$ is monic, to the condition $\gamma^*_{\mathcal{E}/E}(u) = \gamma^*_{\mathcal{E}/E}(v)$, which is equivalent, if $E \not\cong 0$, to $u = v$ (cf. Lemma 6.3.19).

(c) Condition (ii)(a) of Theorem 6.3.4 follows from condition (ii)(a) of Theorem 6.3.8 if the signature of $\mathbb{T}$ contains at least one constant. Indeed, for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, the interpretation of such a constant in M will be an arrow $1 \to MA$ in $\mathcal{E}$, where A is the sort of the constant; condition (ii)(a) of Theorem 6.3.8 thus provides an epimorphic family satisfying the required property.

(d) If all the $\mathbb{T}$-models in $\mathcal{K}$ are finitely generated as Σ-structures and all the Σ-structure homomorphisms of the form $c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ (where $c \in \mathcal{K}$, M is a $\mathbb{T}$-model in $\mathcal{E}$ and $E \in \mathcal{E}$) are injective if $E \not\cong 0$, a sufficient condition for condition (ii)(b) of Theorem 6.3.4 to hold is that, for any context $\vec{x}$, the sequent $(\top \vdash_{\vec{x}} \bigvee_{\phi(\vec{x}) \in \mathcal{I}_{\mathcal{K}}^{\vec{x}}} \phi(\vec{x}))$ be provable in $\mathbb{T}$, where $\mathcal{I}_{\mathcal{K}}^{\vec{x}}$ is the set of (representatives of $\mathbb{T}$-provable-equivalence classes of) geometric formulae in the context $\vec{x}$ which strongly finitely present a $\mathbb{T}$-model in $\mathcal{K}$ (in the sense of Definition 6.2.5). Indeed, under these hypotheses, for any two Σ-structure homomorphisms $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ and $y : d \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, where c and d are finitely generated Σ-structures and $E \not\cong 0$, the substructure $r_e : e \hookrightarrow \mathrm{Hom}_{\mathcal{E}}(E, M)$ of $\mathrm{Hom}_{\mathcal{E}}(E, M)$ generated by the images of c under x and of d under y is finitely generated, say by elements $\xi_1, \ldots, \xi_n$. By choosing a context $\vec{x}$ of the same length as the number of generators of e, we obtain an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in S, E_i \not\cong 0\}$ in $\mathcal{E}$ with the property that for each $i \in I$ there exists a geometric formula $\phi_i(\vec{x})$ strongly presenting a $\mathbb{T}$-model u_i in $\mathcal{K}$ such that $(\xi_1 \circ e_i, \ldots, \xi_n \circ e_i)$ factors through $[[\vec{x} \,.\, \phi_i]]_M$. For each $i \in I$ we thus have a Σ-structure homomorphism $z_i : u_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ which sends the generators of u_i to the tuple $(\xi_1 \circ e_i, \ldots, \xi_n \circ e_i)$. Since z_i is injective by our hypothesis and its image contains a set of generators for e, we have a factorization of $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ r_e$ through z_i. Therefore both $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ x$ and $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ y$ factor through z_i. This shows that condition (ii)(b) is satisfied.

Let us suppose that every $\mathbb{T}$-model in $\mathcal{K}$ is strongly finitely presented by a formula over Σ (in the sense of Definition 6.2.5) and that every $\mathbb{T}$-model homomorphism between two models in $\mathcal{K}$ is induced by a $\mathbb{T}$-provably functional formula between formulae which (strongly) present them. Then, in light of the discussion preceding Definition 6.2.5, we can express the conditions for the functor H_M to be flat (equivalently, filtering) in terms of the satisfaction in M of certain geometric sequents involving these formulae. Specifically, we have the following result:

Theorem 6.3.6 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**) *and $\mathcal{P}$ a set of geometric formulae over Σ such that every $\mathbb{T}$-model in $\mathcal{K}$ is strongly presented by a formula in $\mathcal{P}$ and for any two formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$ presenting respectively models c and d in $\mathcal{K}$, any $\mathbb{T}$-model homomorphism $d \to c$ is induced by a $\mathbb{T}$-provably functional formula from $\phi(\vec{x})$ to $\psi(\vec{y})$. Then $\mathbb{T}$ satisfies condition (i) of Theorem 6.3.1 with respect* *to the category $\mathcal{K}$ if and only if the following conditions are satisfied for every $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$:*

(i) *The sequent*

$$\left(\top \vdash_{[]} \bigvee_{\phi(\vec{x}) \in \mathcal{P}} (\exists \vec{x})\phi(\vec{x}) \right)$$

is valid in M.

(ii) *For any formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$, the sequent*

$$\left(\phi(\vec{x}) \wedge \psi(\vec{y}) \vdash_{\vec{x},\vec{y}} \bigvee_{\substack{\chi(\vec{z}) \in \mathcal{P}, \\ \{\vec{x}\,.\,\phi\} \xleftarrow{[\theta_1]} \{\vec{z}\,.\,\chi\} \xrightarrow{[\theta_2]} \{\vec{y}\,.\,\psi\} \text{ in } \mathcal{C}_{\mathbb{T}}}} (\exists \vec{z})(\theta_1(\vec{z},\vec{x}) \wedge \theta_2(\vec{z},\vec{y})) \right)$$

is valid in M.

(iii) *For any $\mathbb{T}$-provably functional formulae $\theta_1, \theta_2 : \phi(\vec{x}) \to \psi(\vec{y})$ between two formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$, the sequent*

$$\left(\theta_1(\vec{x},\vec{y}) \wedge \theta_2(\vec{x},\vec{y}) \vdash_{\vec{x},\vec{y}} \bigvee_{\substack{\chi(\vec{z}) \in \mathcal{P}, \\ \{\vec{z}\,.\,\chi\} \xrightarrow{[\tau]} \{\vec{x}\,.\,\phi\} \text{ in } \mathcal{C}_{\mathbb{T}} \mid \\ \tau \wedge \theta_1 \dashv\vdash_{\mathbb{T}} \tau \wedge \theta_2}} (\exists \vec{z})\tau(\vec{z},\vec{x}) \right)$$

is valid in M.

Proof The fact that any model c in $\mathcal{K}$ is strongly finitely presented by a formula $\phi(\vec{x})$ in $\mathcal{P}$ ensures that for any object E of $\mathcal{E}$ and any $\mathbb{T}$-model M in $\mathcal{E}$, the Σ-homomorphisms $c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ are in bijective correspondence with the generalized elements $E \to [[\vec{x}\,.\,\phi]]_M$ in $\mathcal{E}$. Our thesis thus follows from the Kripke-Joyal semantics for the topos $\mathcal{E}$, noticing that, by our hypotheses, for any two formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$ presenting respectively models c and d in $\mathcal{K}$, any $\mathbb{T}$-model homomorphism $d \to c$ is induced by a $\mathbb{T}$-provably functional formula from $\phi(\vec{x})$ to $\psi(\vec{y})$. □

Remarks 6.3.7 (a) Under the hypotheses of Theorem 6.3.6, all the sequents in the statement of the theorem are satisfied by every model in $\mathcal{K}$. Therefore, adding them to $\mathbb{T}$ yields a quotient of $\mathbb{T}$ satisfying condition (i) of Theorem 6.3.1 with respect to the category $\mathcal{K}$.

(b) Under the alternative hypothesis that every model of $\mathcal{K}$ is both strongly finitely presentable and finitely generated (with respect to the same generators), for any two formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$ presenting respectively models c and d in $\mathcal{K}$, the $\mathbb{T}$-model homomorphisms $c \to d$ are in bijection (via the evaluation of such homomorphisms at the generators of c) with the tuples of elements of d which satisfy ϕ, each of which has the form $(t_1(\vec{y}), \ldots, t_n(\vec{y}))$ (where n is the length of the context $\vec{x}$) for some terms $t_1, \ldots, t_n$ in the context $\vec{y}$ over the signature Σ. Conditions (ii) and (iii) of Theorem 6.3.6 can thus be reformulated more explicitly as follows (using the notation of section 6.1.4):

(ii') For any formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$, where $\vec{x} = (x_1^{A_1}, \ldots, x_n^{A_n})$ and $\vec{y} = (y_1^{B_1}, \ldots, y_m^{B_m})$, the sequent

$$\left(\phi(\vec{x}) \wedge \psi(\vec{y}) \vdash_{\vec{x},\vec{y}} \bigvee_{\substack{\chi(\vec{z}) \in \mathcal{P},\ t_1^{A_1}(\vec{z}), \ldots, t_n^{A_n}(\vec{z}) \\ s_1^{B_1}(\vec{z}), \ldots, s_m^{B_m}(\vec{z})}} (\exists \vec{z}) \left(\chi(\vec{z}) \wedge \bigwedge_{\substack{i \in \{1, \ldots, n\}, \\ j \in \{1, \ldots, m\}}} (x_i = t_i(\vec{z}) \wedge y_j = s_j(\vec{z}))\right)\right),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ in $\mathcal{P}$ and all the tuples of terms $t_1^{A_1}(\vec{z}), \ldots, t_n^{A_n}(\vec{z})$ and $s_1^{B_1}(\vec{z}), \ldots, s_m^{B_m}(\vec{z})$ such that

$$(t_1^{A_1}(\vec{\xi_\chi}), \ldots, t_n^{A_n}(\vec{\xi_\chi})) \in [[\vec{x} \,.\, \phi]]_{M_{\{\vec{z} \,.\, \chi\}}}$$

and

$$(s_1^{B_1}(\vec{\xi_\chi}), \ldots, s_m^{B_m}(\vec{\xi_\chi})) \in [[\vec{y} \,.\, \psi]]_{M_{\{\vec{z} \,.\, \chi\}}},$$

is valid in M.

(iii') For any formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ in $\mathcal{P}$, where $\vec{x} = (x_1^{A_1}, \ldots, x_n^{A_n})$ and $\vec{y} = (y_1^{B_1}, \ldots, y_m^{B_m})$, and any terms $t_1^{A_1}(\vec{y}), s_1^{A_1}(\vec{y}), \ldots, t_n^{A_n}(\vec{y})$, $s_n^{A_n}(\vec{y})$ such that

$$(t_1(\vec{\xi_\phi}), \ldots, t_n(\vec{\xi_\phi})) \in [[\vec{y} \,.\, \phi]]_{M_{\{\vec{y} \,.\, \psi\}}}$$

and

$$(s_1(\vec{\xi_\phi}), \ldots, s_n(\vec{\xi_\phi})) \in [[\vec{y} \,.\, \phi]]_{M_{\{\vec{y} \,.\, \psi\}}},$$

the sequent

$$\left(\bigwedge_{i \in \{1, \ldots, n\}} (t_i(\vec{y}) = s_i(\vec{y})) \wedge \psi(\vec{y}) \vdash_{\vec{y}} \bigvee_{\chi(\vec{z}) \in \mathcal{P},\, u_1^{B_1}(\vec{z}), \ldots, u_m^{B_m}(\vec{z})} (\exists \vec{z}) \left(\chi(\vec{z}) \wedge \bigwedge_{j \in \{1, \ldots, m\}} (y_j = u_j(\vec{z}))\right)\right),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ in $\mathcal{P}$ and all the tuples of terms $u_1^{B_1}(\vec{z}), \ldots, u_m^{B_m}(\vec{z})$ such that

$$(u_1(\vec{\xi_\chi}), \dots, u_m(\vec{\xi_\chi})) \in [[\vec{y} \,.\, \psi]]_{M_{\{\vec{z}\,.\,\chi\}}}$$

and

$$t_i(u_1(\vec{\xi_\chi}), \dots, u_m(\vec{\xi_\chi})) = s_i(u_1(\vec{\xi_\chi}), \dots, u_m(\vec{\xi_\chi})) \text{ in } M_{\{\vec{z}\,.\,\chi\}}$$

for all $i \in \{1, \dots, n\}$, is valid in M.

Condition (ii)

In this section we shall give more concrete reformulations of condition (ii) of Theorem 6.3.1.

Let $\mathcal{E}$ be a Grothendieck topos with a separating family of objects $\mathcal{S}$ and M a $\mathbb{T}$-model in $\mathcal{E}$ satisfying condition (i) of Theorem 6.3.1, i.e. such that the functor H_M is flat. The canonical morphism $\widetilde{H_M}(M_{\mathbb{T}}) \to M$ considered in condition (ii) is an isomorphism if and only if, for every sort A of Σ, the induced arrow $\widetilde{H_M}(\{x^A \,.\, \top\}) \to MA$ is an isomorphism in $\mathcal{E}$.

By Theorem 5.2.5, we have that $\widetilde{H_M}(\{x^A \,.\, \top\}) \cong \mathrm{colim}(H_M \circ \pi^{\mathcal{K}}_{\{x^A\,.\,\top\}})$, where $\pi^{\mathcal{K}}_{\{x^A\,.\,\top\}}$ is the projection functor $\mathcal{A}^{\mathcal{K}}_{\{x^A\,.\,\top\}} \to \mathcal{K}^{\mathrm{op}}$ from the category $\mathcal{A}^{\mathcal{K}}_{\{x^A\,.\,\top\}}$ having as objects the pairs (c, z), where c is a $\mathbb{T}$-model in $\mathcal{K}$ and z is an element of the set cA, and as arrows $(c, z) \to (d, w)$ the $\mathbb{T}$-model homomorphisms $g : d \to c$ in $\mathcal{K}$ such that $g_A(w) = z$. Proposition 5.2.8 provides the following explicit representation of $\mathrm{colim}(H_M \circ \pi^{\mathcal{K}}_{\{x^A\,.\,\top\}})$ as the quotient

$$\left(\coprod_{(c,z)\in\mathcal{A}^{\mathcal{K}}_{\{x^A\,.\,\top\}}} H_M(c) \right) / R_{\{x^A\,.\,\top\}},$$

where $R_{\{x^A\,.\,\top\}}$ is the equivalence relation defined by the condition that, for any objects (c, z), (d, w) of $\mathcal{A}^{\mathcal{K}}_{\{x^A\,.\,\top\}}$ and any generalized elements $x : E \to F(c)$ and $x' : E \to F(d)$, $(x, x') \in R_{\{x^A\,.\,\top\}}$ if and only if there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model a_i in $\mathcal{K}$, a generalized element $h_i : E_i \to F(b_i)$ and two $\mathbb{T}$-model homomorphisms $f_i : c \to b_i$ and $f'_i : d \to b_i$ in $\mathcal{K}$ such that ${f_i}_A(z) = {f'_i}_A(w)$ and $< F(f_i), F(f'_i) > \circ h_i = < x, x' > \circ e_i$. From this representation it follows that the canonical arrow $\widetilde{H_M}(\{x^A \,.\, \top\}) \to MA$ is an isomorphism if and only if (using the notation of Proposition 5.2.8)

(1) the canonical arrows $\kappa_{(c,z)} : H_M(c) \to MA$ for $(c, z) \in \mathcal{A}_{\{x^A\,.\,\top\}}$ are jointly epimorphic and
(2) for any two objects (c, z) and (d, w) of the category $\mathcal{A}_{\{x^A\,.\,\top\}}$ and any generalized elements $x : E \to H_M(c)$ and $x' : E \to H_M(d)$ (where $E \in \mathcal{S}$), $\kappa_{(c,z)} \circ x = \kappa_{(d,w)} \circ x'$ if and only if $(x, x') \in R_{\{x^A\,.\,\top\}}$.

Thanks to the identification between the generalized elements $x : E \to H_M(c)$ and the Σ-structure homomorphisms $f_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ provided by Proposition 6.2.3, we can rewrite conditions (1) and (2) more explicitly, as follows.

We preliminarily notice that, for any object (c, z) of the category $\mathcal{A}_{\{x^A . \top\}}$, the canonical arrow $\kappa_{(c,z)} : H_M(c) = \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M) \to MA$ can be described in terms of generalized elements as the arrow sending any generalized element $x : E \to H_M(c)$, corresponding via the identification of Proposition 6.2.3 to a Σ-structure homomorphism $f_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, to the generalized element $E \to MA$ of MA given by $(f_x)_A(z)$. Therefore, for any generalized elements $x : E \to H_M(c)$ and $x' : E \to H_M(d)$ corresponding respectively to Σ-structure homomorphisms $f_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ and $f_{x'} : d \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, we have that $\kappa_{(c,z)} \circ x = \kappa_{(d,w)} \circ x'$ if and only if $(f_x)_A(z) = (f_{x'})_A(w)$.

Now, condition (1) can be formulated by saying that for any generalized element $x : E \to MA$ (where $E \in \mathcal{S}$) there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$, a family $\{(c_i, z_i) \mid i \in I\}$ of objects of the category $\mathcal{A}_{\{x^A . \top\}}$ and generalized elements $y_i : E_i \to H_M(c_i)$ (for each $i \in I$) such that $x \circ e_i = \kappa_{(c_i,z_i)} \circ y_i$. By Proposition 6.2.3, it thus rewrites as follows: for any generalized element $x : E \to MA$ (where $E \in \mathcal{S}$), there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$, a family $\{(c_i, z_i) \mid i \in I\}$ of objects of the category $\mathcal{A}_{\{x^A . \top\}}$ and Σ-structure homomorphisms $f_i : c_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ (for each $i \in I$) such that $(f_i)_A(z_i) = x \circ e_i$.

Concerning condition (2), we observe that, for any objects (c, z) and (d, w) of $\mathcal{A}_{\{x^A . \top\}}$ and any Σ-structure homomorphisms $f_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ and $f_{x'} : d \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ (where $E \in \mathcal{S}$) corresponding respectively to generalized elements $x : E \to H_M(c)$ and $x' : E \to H_M(d)$ via the identification of Proposition 6.2.3, we have that $(x, x') \in R_{\{x^A . \top\}}$ if and only if there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model b_i in $\mathcal{K}$, a Σ-structure homomorphism $h_i : b_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ and two $\mathbb{T}$-model homomorphisms $f_i : c \to b_i$ and $f'_i : d \to b_i$ in $\mathcal{K}$ such that $(f_i)_A(z) = (f'_i)_A(w)$ and $h_i \circ f_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ f_x$ and $h_i \circ f'_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ f_{x'}$.

Summarizing, we have the following result:

Theorem 6.3.8 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**)*, $\mathcal{E}$ a Grothendieck topos with a separating family of objects $\mathcal{S}$ and M a $\mathbb{T}$-model in $\mathcal{E}$. Then, under the assumption that $\mathbb{T}$ satisfies condition (i) of Theorem 6.3.1 with respect to $\mathcal{K}$, the following conditions are equivalent:*

(i) *The extension $\widetilde{H_M} : \mathcal{C}_{\mathbb{T}} \to \mathcal{E}$ of the functor $H_M : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$ to the syntactic category $\mathcal{C}_{\mathbb{T}}$ (in the sense of section 5.2.3) satisfies the property that the canonical morphism $\widetilde{H_M}(M_{\mathbb{T}}) \to M$ is an isomorphism.*

(ii) *For every sort A of Σ, the following conditions are satisfied:*

  (a) *For any generalized element $x : E \to MA$ (where $E \in \mathcal{S}$), there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model c_i in $\mathcal{K}$, an element z_i of c_iA and a Σ-structure homomorphism $f_i : c_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $(f_i)_A(z_i) = x \circ e_i$.*

(b) *For any two pairs* (c, z) *and* (d, w)*, where* c *and* d *are* $\mathbb{T}$*-models in* $\mathcal{K}$ *and* $z \in cA, w \in dA$*, and any* Σ*-structure homomorphisms* $f : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ *and* $f' : d \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ *(where* $E \in \mathcal{S}$*), we have that* $f_A(z) = f'_A(w)$ *if and only if there exist an epimorphic family* $\{e_j : E_j \to E \mid j \in J, E_j \in \mathcal{S}\}$ *in* $\mathcal{E}$ *and for each* $j \in J$ *a* $\mathbb{T}$*-model* b_j *in* $\mathcal{K}$*, a* Σ*-structure homomorphism* $h_j : b_j \to \mathrm{Hom}_{\mathcal{E}}(E_j, M)$ *and two* $\mathbb{T}$*-model homomorphisms* $f_j : c \to b_j$ *and* $f'_j : d \to b_j$ *in* $\mathcal{K}$ *such that* $(f_j)_A(z) = (f_j)_{A'}(w)$*,* $h_j \circ f_j = \mathrm{Hom}_{\mathcal{E}}(e_j, M) \circ f$ *and* $h_j \circ f'_j = \mathrm{Hom}_{\mathcal{E}}(e_j, M) \circ f'$*.*

Remarks 6.3.9 (a) If all the $\mathbb{T}$-model homomorphisms in any Grothendieck topos are monic and $\mathbb{T}$ satisfies condition (ii)(b) of Theorem 6.3.4 then condition (ii)(b) of Theorem 6.3.8 is automatically satisfied.

(b) A sufficient condition for condition (ii)(a) of Theorem 6.3.8 to hold is that the sequent $\left(\top \vdash_x \bigvee_{\phi(x) \in \mathcal{I}_{\mathcal{K}}^x} \phi(x)\right)$ be provable in $\mathbb{T}$, where $\mathcal{I}_{\mathcal{K}}^x$ is the set of (representatives of $\mathbb{T}$-provable-equivalence classes of) geometric formulae in one variable which strongly finitely present a $\mathbb{T}$-model in $\mathcal{K}$ (in the sense of Definition 6.2.5).

The following result provides an explicit reformulation of condition (ii) of Theorem 6.3.1 holding for theories $\mathbb{T}$ with respect to full subcategories $\mathcal{K} \hookrightarrow$ f.p.$\mathbb{T}$-mod(**Set**) such that every model of $\mathcal{K}$ is both strongly finitely presentable and finitely generated (with respect to the same generators).

Theorem 6.3.10 *Let* $\mathbb{T}$ *be a geometric theory over a signature* Σ *and* $\mathcal{K}$ *a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**)*. If every model in* $\mathcal{K}$ *is both strongly finitely presentable and finitely generated (with respect to the same generators) then* $\mathbb{T}$ *satisfies condition (ii) of Theorem 6.3.1 with respect to the category* $\mathcal{K}$ *if and only if for every model* M *of* $\mathbb{T}$ *in a Grothendieck topos satisfying condition (i) of Theorem 6.3.1 (i.e. such that the functor* H_M *is flat), the following sequents are satisfied in* M *(where* $\mathcal{P}$ *denotes the set of (representatives of) formulae over* Σ *which present a model in* $\mathcal{K}$ *and the notation is that of section 6.1.4):*

(i) *for any sort* A *of* Σ*, the sequent*

$$\left(\top \vdash_{x_A} \bigvee_{\chi(\vec{z}) \in \mathcal{P},\, t^A(\vec{z})} (\exists \vec{z})(\chi(\vec{z}) \wedge x = t(\vec{z}))\right),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ *in* $\mathcal{P}$ *and all the terms* $t^A(\vec{z})$*;*

(ii) *for any sort* A *of* Σ*, any formulae* $\phi(\vec{x})$ *and* $\psi(\vec{y})$ *in* $\mathcal{P}$*, where* $\vec{x} = (x_1^{A_1}, \ldots, x_n^{A_n})$ *and* $\vec{y} = (y_1^{B_1}, \ldots, y_m^{B_m})$*, and any terms* $t^A(\vec{x})$ *and* $s^A(\vec{y})$*, the sequent*

$$\Big(\phi(\vec{x}) \wedge \psi(\vec{y}) \wedge t(\vec{x}) = s(\vec{y}) \vdash_{\vec{x},\vec{y}}$$

$$\bigvee_{\substack{\chi(\vec{z})\in\mathcal{P},\ p_1^{A_1}(\vec{z}),\dots,p_n^{A_n}(\vec{z})\\ q_1^{B_1}(\vec{z}),\dots,q_m^{B_m}(\vec{z})}} (\exists \vec{z})\Big(\chi(\vec{z}) \wedge \bigwedge_{\substack{i\in\{1,\dots,n\},\\ j\in\{1,\dots,m\}}} (x_i = p_i(\vec{z}) \wedge y_j = q_j(\vec{z}))\Big)\Big),$$

where the disjunction is taken over all the formulae $\chi(\vec{z})$ in $\mathcal{P}$ and all the tuples of terms $p_1^{A_1}(\vec{z}),\dots,p_n^{A_n}(\vec{z})$ and $q_1^{B_1}(\vec{z}),\dots,q_m^{B_m}(\vec{z})$ such that $(p_1(\vec{\xi_\chi}),\dots,p_n(\vec{\xi_\chi})) \in [[\vec{x}\,.\,\phi]]_{M_{\{\vec{z}\,.\,\chi\}}}$, $(q_1(\vec{\xi}),\dots,q_m(\vec{\xi_\chi})) \in [[\vec{y}\,.\,\psi]]_{M_{\{\vec{z}\,.\,\chi\}}}$ and $t(p_1(\vec{\xi_\chi}),\dots,p_n(\vec{\xi_\chi})) = s(q_1(\vec{\xi_\chi}),\dots,q_m(\vec{\xi_\chi}))$ in $M_{\{\vec{z}\,.\,\chi\}}$.

Proof The thesis follows from Theorem 6.3.8 in light of the Kripke-Joyal semantics and the fact that for every model c in $\mathcal{K}$ strongly finitely presented by a formula $\phi(\vec{x})$ in $\mathcal{P}$, any object E of $\mathcal{E}$ and any $\mathbb{T}$-model M in $\mathcal{E}$, the Σ-homomorphisms $c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ are in bijective correspondence with the generalized elements $E \to [[\vec{x}\,.\,\phi]]_M$ in $\mathcal{E}$. □

Remark 6.3.11 All the sequents in the statement of Theorem 6.3.10 are satisfied by every $\mathbb{T}$-model in $\mathcal{K}$ (cf. Remark 6.3.2(d)); therefore, adding them to any theory $\mathbb{T}$ satisfying the hypotheses of the theorem and condition (i) of Theorem 6.3.1 yields a quotient of $\mathbb{T}$ satisfying condition (ii) of Theorem 6.3.1 with respect to the category $\mathcal{K}$.

Condition (iii)

In this section, we shall give 'concrete' reformulations of conditions (iii)(a)-(b)-(c) of Theorem 6.3.1.

The following proposition shows that, under some natural assumptions which are often verified in practice, condition (iii)(a) of Theorem 6.3.1 is satisfied.

Proposition 6.3.12 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**)*.*

(i) *If $\mathbb{T}$ is a quotient of a theory $\mathbb{S}$ satisfying condition (iii)(a) of Theorem 6.3.1 with respect to a full subcategory $\mathcal{H}$ of* f.p.$\mathbb{S}$-mod(**Set**) *and $\mathcal{K}$ is a subcategory of $\mathcal{H}$ then $\mathbb{T}$ satisfies condition (iii)(a) of Theorem 6.3.1 with respect to the category $\mathcal{K}$.*

(ii) *If every $\mathbb{T}$-model in $\mathcal{K}$ is strongly finitely presented (in the sense of section 6.2.2) then $\mathbb{T}$ satisfies condition (iii)(a) of Theorem 6.3.1 with respect to the category $\mathcal{K}$.*

(iii) *If for every $\mathbb{T}$-model c in $\mathcal{K}$ the object $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$ can be built from c and M by only using geometric constructions (i.e. constructions only involving finite limits and small colimits) then $\mathbb{T}$ satisfies condition (iii)(a) of Theorem 6.3.1 with respect to the category $\mathcal{K}$.*

Proof

(i) If $\mathbb{T}$ is a quotient of $\mathbb{S}$ then, for every Grothendieck topos $\mathcal{E}$, the category $\mathbb{T}\text{-mod}(\mathcal{E})$ is a full subcategory of the category $\mathbb{S}\text{-mod}(\mathcal{E})$. This clearly implies that for any $\mathbb{T}$-models M and N in a Grothendieck topos $\mathcal{E}$, we have

$$\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(M, N) \cong \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{S}\text{-mod}(\mathcal{E})}}(M, N);$$

in particular, for any $\mathbb{T}$-model c in $\mathcal{K}$ and any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M) \cong \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{S}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M)$. The fact that $\mathbb{S}$ satisfies condition (iii)(a) of Theorem 6.3.1 with respect to the category $\mathcal{H}$ thus implies that $\mathbb{T}$ does so with respect to the category $\mathcal{K}$, as required.

(ii) If every $\mathbb{T}$-model c in $\mathcal{K}$ is strongly finitely presented by a formula $\phi(\vec{x})$ over the signature of $\mathbb{T}$ then, for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, $\mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), M) \cong [[\vec{x}\,.\,\phi]]_M$ (cf. section 6.2.2). Therefore, as the interpretation of geometric formulae is preserved by inverse image functors of geometric morphisms, condition (iii)(a) of Theorem 6.3.1 is satisfied by the theory $\mathbb{T}$ with respect to the category $\mathcal{K}$.

(iii) Condition (iii) of Proposition 6.3.12 implies condition (iii)(a) of Theorem 6.3.1 since geometric constructions are preserved by inverse image functors of geometric morphisms.

□

We shall now proceed to give concrete reformulations of conditions (iii)(b)-(1) and (iii)(b)-(2) of Remark 6.3.2(f), in order to make them more easily verifiable in practice.

First, let us explicitly describe, for any object c of $\mathcal{K}$, the arrow

$$\eta^F(c) : F(c) \to \mathrm{Hom}^{\mathcal{E}}_{\underline{\mathbb{T}\text{-mod}(\mathcal{E})}}(\gamma^*_{\mathcal{E}}(c), \tilde{F}(M_{\mathbb{T}}))$$

of condition (iii)(b) of Theorem 6.3.1 in terms of generalized elements.

For any generalized element $x : E \to F(c)$, $\eta^F(c) \circ x$ corresponds under the identification of Proposition 6.2.3 to the Σ-structure homomorphism $z_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$ defined at each sort A of Σ as the function $cA \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(\{x^A\,.\,\top\}))$ sending any element $y \in cA$ to the generalized element $E \to \tilde{F}(\{x^A\,.\,\top\})$ obtained by composing the canonical colimit arrow $\kappa^F_{(c,y)} : F(c) \to \tilde{F}(\{x^A\,.\,\top\})$ with the generalized element $x : E \to F(c)$. Given a separating family of objects $\mathcal{S}$ for $\mathcal{E}$, it follows that $\eta^F(c)$ is a monomorphism if and only if for any generalized elements $x, x' : E \to F(c)$ (where $E \in \mathcal{S}$), $\kappa^F_{(c,y)} \circ x = \kappa^F_{(c,y)} \circ x'$ for every sort A of Σ and every element $y \in cA$ implies $x = x'$. By Proposition 5.2.8 we have that $\kappa^F_{(c,y)} \circ x = \kappa^F_{(c,y)} \circ x'$ if and only if there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I,\ E_i \in S\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model a_i in $\mathcal{K}$, a generalized element $h_i : E_i \to F(a_i)$ and two

$\mathbb{T}$-model homomorphisms $f_i, f'_i : c \to a_i$ in $\mathcal{K}$ such that $(f_i)_A(y) = (f'_i)_A(y)$ and $< F(f_i), F(f'_i) > \circ h_i =< x, x' > \circ e_i$.

To obtain an explicit characterization of the condition for $\eta^F(c)$ to be an epimorphism, we notice that an arrow $f : A \to B$ in a Grothendieck topos $\mathcal{E}$ with a separating family of objects $\mathcal{S}$ is an epimorphism if and only if for every generalized element $x : E \to B$ of B (where $E \in \mathcal{S}$) there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I,\ E_i \in S\}$ in $\mathcal{E}$ and for each $i \in I$ a generalized element $y_i : E_i \to A$ such that $f \circ y_i = x \circ e_i$. Applying this characterization to the arrow $\eta^F(c)$ in light of Proposition 6.2.3, we obtain the following criterion: $\eta^F(c)$ is an epimorphism if and only if for every object E of $\mathcal{E}$ and any Σ-structure homomorphism $v : c \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I,\ E_i \in S\}$ in $\mathcal{E}$ and for each $i \in I$ a generalized element $x_i : E_i \to F(c)$ such that $z_{x_i} = \mathrm{Hom}_{\mathcal{E}}(e_i, \tilde{F}(M_{\mathbb{T}})) \circ v$ for all i (where $z_{x_i} = \xi_{\eta^F(c) \circ x_i}$ using the notation of Proposition 6.2.3). The condition '$z_{x_i} = \mathrm{Hom}_{\mathcal{E}}(e_i, \tilde{F}(M_{\mathbb{T}})) \circ v$' can be explicitly reformulated as the requirement that, for every sort A of Σ and every element $y \in cA$, $\kappa^F_{(c,y)} \circ x_i = v_A(y) \circ e_i$.

Summarizing, we have the following result:

Theorem 6.3.13 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**)*, $\mathcal{E}$ a Grothendieck topos with a separating family of objects $\mathcal{S}$ and $F : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$ a flat functor. Then*

(i) *F satisfies condition (iii)(b)-(1) of Remark 6.3.2(f) if and only if for any $\mathbb{T}$-model c in $\mathcal{K}$ and any generalized elements $x, x' : E \to F(c)$ (where $E \in \mathcal{S}$), if for every sort A of Σ and every element $y \in cA$ there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model a_i in $\mathcal{K}$, a generalized element $h_i : E_i \to F(a_i)$ and two $\mathbb{T}$-model homomorphisms $f_i, f'_i : c \to a_i$ such that $(f_i)_A(y) = (f'_i)_A(y)$ and $< F(f_i), F(f'_i) > \circ h_i =< x, x' > \circ e_i$ then $x = x'$.*

(ii) *F satisfies condition (iii)(b)-(2) of Remark 6.3.2(f) if and only if for any $\mathbb{T}$-model c in $\mathcal{K}$, any object E of $\mathcal{S}$ and any Σ-structure homomorphism $v : c \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I, E_i \in \mathcal{S}\}$ in $\mathcal{E}$ and for each $i \in I$ a generalized element $x_i : E_i \to F(c)$ such that, for every sort A of Σ and every element $y \in cA$, $\kappa^F_{(c,y)} \circ x_i = v_A(y) \circ e_i$.* □

Under the hypothesis that for every sort A of Σ the formula $\{x^A . \top\}$ presents a $\mathbb{T}$-model F_A in $\mathcal{K}$, we have that $\tilde{F}(M_{\mathbb{T}})A = \tilde{F}(\{x^A . \top\}) \cong F(F_A)$, and the homomorphism $z_x : c \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$ corresponding to a generalized element $x : E \to F(c)$ can be described as follows: for any sort A of Σ, $(z_x)_A : cA \to \mathrm{Hom}_{\mathcal{E}}(E, F(F_A))$ assigns to any element $y \in cA$, corresponding to a $\mathbb{T}$-model homomorphism $s_y : F_A \to c$ via the universal property of F_A, the generalized element $F(s_y) \circ x$. Therefore, for any generalized elements $x, x' : E \to F(c)$ and any element $y \in cA$, $(z_x)_A(y) = (z_{x'})_A(y)$ if and only if $F(s_y) \circ x = F(s_y) \circ x'$.

This gives the following result:

Proposition 6.3.14 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**) *and $F : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$ a flat functor with values in a Grothendieck topos $\mathcal{E}$. Suppose that for every sort A of Σ the formula $\{x^A . \top\}$ presents a $\mathbb{T}$-model F_A and that all the models F_A lie in $\mathcal{K}$. Then F satisfies condition (iii)(b)-(1) of Remark 6.3.2(f) with respect to the category $\mathcal{K}$ if and only if for every $\mathbb{T}$-model c in $\mathcal{K}$ the family of arrows $\{F(s_y) : F(c) \to F(F_A) \mid A \in \Sigma, y \in cA\}$ is jointly monic.* □

The characterization of $\tilde{F}(M_{\mathbb{T}})$ as an $\mathcal{E}$-filtered $\mathcal{E}$-indexed colimit established in section 5.2.3 allows us to obtain a different reformulation, in terms of the Σ-structure homomorphisms $\xi_{(c,x)} : F(c) \to \tilde{F}(M_{\mathbb{T}})$ defined in that context, of Theorem 6.3.13, as follows.

Theorem 6.3.15 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**) *and $F : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$ a flat functor with values in a Grothendieck topos $\mathcal{E}$. Then*

(i) *F satisfies condition (iii)(b)-(1) of Remark 6.3.2(f) if and only if for any $\mathbb{T}$-model c in $\mathcal{K}$ and any generalized elements $x, x' : E \to F(c)$, the Σ-structure homomorphisms $\xi_{(c,x)}$ and $\xi_{(c,x')}$ are equal if and only if $x = x'$.*

(ii) *F satisfies condition (iii)(b)-(2) of Remark 6.3.2(f) if and only if for any $\mathbb{T}$-model c in $\mathcal{K}$, any object E of $\mathcal{E}$ and any Σ-structure homomorphism $v : c \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a generalized element $x_i : E_i \to F(c)$ such that $\mathrm{Hom}_{\mathcal{E}}(e_i, \tilde{F}(M_{\mathbb{T}})) \circ v = \xi_{(c,x_i)}$.*

Proof

(i) The equality $\xi_{(c,x)} = \xi_{(c,x')}$ holds if and only if, for every sort A of Σ and every $y \in cA$, ${\xi_{(c,x)}}_A(y) = {\xi_{(c,x')}}_A(y)$. But by Proposition 5.2.11 this latter condition is satisfied if and only if there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model a_i in $\mathcal{K}$, a generalized element $h_i : E_i \to F(a_i)$ and two $\mathbb{T}$-model homomorphisms $f_i, f'_i : c \to a_i$ such that $(f_i)_A(y) = (f'_i)_A(y)$ and $< F(f_i), F(f'_i) > \circ h_i = < x, x' > \circ e_i$. Our thesis thus follows from Theorem 6.3.13(i).

(ii) By Theorem 6.3.13(ii), it suffices to notice that, for every sort A of Σ and every element $y \in cA$, the condition '$\kappa^F_{(c,y)} \circ x_i = v_A(y) \circ e_i$' can be reformulated as the requirement that $\mathrm{Hom}_{\mathcal{E}}(e_i, \tilde{F}(M_{\mathbb{T}})) \circ v = \xi_{(c,x_i)}$ (for any $i \in I$). □

Thanks to this different reformulation of condition (iii)(b)-(1) of Remark 6.3.2(f), we can prove that if all the arrows in $\mathcal{K}$ are sortwise injective then $\mathbb{T}$ satisfies this condition.

First, we need a lemma.

Lemma 6.3.16 *Under the hypotheses of Theorem 6.3.15, if all the homomorphisms in $\mathcal{K}$ are sortwise injective then, for any pair (c, x) consisting of an object c of $\mathcal{K}$ and a generalized element $x : E \to F(c)$ such that $E \not\cong 0_{\mathcal{E}}$, the Σ-structure homomorphism $\xi_{(c,x)} : c \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$ is sortwise injective.*

Proof For any sort A of Σ, the function ${\xi_{(c,x)}}_A : cA \to \mathrm{Hom}_{\mathcal{E}}(E, MA)$ sends any element $y \in cA$ to the generalized element $\kappa_{(c,y)} \circ x$. Now, for any $y_1, y_2 \in cA$, we have, by Proposition 5.2.8, that $\kappa_{(c,y_1)} \circ x = \kappa_{(c,y_2)} \circ x$ if and only if there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{T}$-model a_i in $\mathcal{K}$, a generalized element $h_i : E_i \to F(a_i)$ and a $\mathbb{T}$-model homomorphism $f_i : c \to a_i$ in $\mathcal{K}$ such that $(f_i)_A(y_1) = (f_i)_A(y_2)$ and $F(f_i) \circ h_i = x \circ e_i$. If $E \not\cong 0_{\mathcal{E}}$ then the set I is non-empty; given $i \in I$, we thus have that $(f_i)_A(y_1) = (f_i)_A(y_2)$, which entails $y_1 = y_2$ as f_i is sortwise injective by our hypothesis. □

Corollary 6.3.17 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**) *whose arrows are all sortwise injective. Then $\mathbb{T}$ satisfies condition (iii)(b)-(1) of Remark 6.3.2(f) with respect to the category $\mathcal{K}$.*

Proof By Proposition 5.2.11, for any flat functor $F : \mathcal{K}^{\mathrm{op}} \to \mathcal{E}$ and any generalized elements $x : E \to F(c)$ and $x' : E \to F(c)$, the following 'joint embedding property' holds: there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$, where we suppose without loss of generality all the objects E_i to be non-zero, and for each $i \in I$ a $\mathbb{T}$-model c_i in $\mathcal{K}$, $\mathbb{T}$-model homomorphisms $f_i, g_i : c \to c_i$ and a generalized element $x_i : E_i \to F(c_i)$ such that $< x, x' > \circ e_i = < F(f_i), F(g_i) > \circ x_i$, $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ \xi_{(c,x)} = \xi_{(c_i,x_i)} \circ f_i$ and $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ \xi_{(c,x')} = \xi_{(c_i,x_i)} \circ g_i$ (for all $i \in I$).

Now, since all the arrows in $\mathcal{K}$ are sortwise injective homomorphisms, by Lemma 6.3.16 the homomorphism $\xi_{(c_i,x_i)}$ is sortwise injective (for each $i \in I$) and hence $\xi_{(c,x)} = \xi_{(c,x')}$ implies $f_i = g_i$. But then $x \circ e_i = F(f_i) \circ x_i = F(g_i) \circ x_i = x' \circ e_i$ for all $i \in I$, whence $x = x'$, as required. □

We shall now proceed to identify some natural sufficient conditions for a theory $\mathbb{T}$ to satisfy condition (iii)(b)-(2) of Remark 6.3.2(f). Before doing so, we need a number of preliminary results.

Lemma 6.3.18 *Let Σ be a signature without relation symbols and $\mathbb{T}$ a geometric theory over a signature Σ' obtained from Σ by solely adding relation symbols whose interpretation in any $\mathbb{T}$-model in a Grothendieck topos is the complement of the interpretation of a geometric formula over Σ (for instance, $\mathbb{T}$ can be the injectivization of a geometric theory over Σ in the sense of Definition 7.2.10). Let $f : M \to N$ and $g : P \to N$ be homomorphisms of $\mathbb{T}$-models in a Grothendieck topos $\mathcal{E}$, and let k be an assignment to any sort A of Σ of an arrow $k_A : MA \to PA$ such that $g_A \circ k_A = f_A$. Then, if g is sortwise monic, k defines a $\mathbb{T}$-model homomorphism $M \to P$ such that $g \circ k = f$.*

Proof The fact that k preserves the interpretation of function symbols of Σ' follows from the fact that f and g do, as g is sortwise monic. It remains to prove that, for any relation symbol $R \rightarrowtail A_1 \cdots A_n$ of Σ' and any generalized element $x : E \to MA_1 \times \cdots \times MA_n$ in $\mathcal{E}$, if x factors through $MR \rightarrowtail MA_1 \times \cdots \times MA_n$ then $(k_{A_1} \times \cdots \times k_{A_n}) \circ x$ factors through $PR \rightarrowtail PA_1 \times \cdots \times PA_n$. Now, if PR is the complement of the interpretation $i_{\{\vec{x}\,.\,\phi\}} : [[\vec{x}\,.\,\phi]]_P \rightarrowtail PA_1 \times \cdots \times PA_n$ of a geometric formula $\phi(\vec{x}) = \phi(x^{A_1}, \ldots, x^{A_n})$ in the model P, this latter condition is equivalent to the requirement that the equalizer e of $(k_{A_1} \times \cdots \times k_{A_n}) \circ x$ and $i_{\{\vec{x}\,.\,\phi\}}$ be zero; but, since g is a Σ'-structure homomorphism, the subobject e is contained in the equalizer of the arrows $(g_{A_1} \times \cdots \times g_{A_n}) \circ (k_{A_1} \times \cdots \times k_{A_n}) \circ x = (f_{A_1} \times \cdots \times f_{A_n}) \circ x$ and $[[\vec{x}\,.\,\phi]]_N \rightarrowtail NA_1 \times \cdots \times NA_n$, which is zero since f is a Σ'-structure homomorphism and NR is the complement of $[[\vec{x}\,.\,\phi]]_N$. Therefore e is zero, as required. □

Lemma 6.3.19 *Let $\mathcal{E}$ be a Grothendieck topos. Then the inverse image functor $\gamma_{\mathcal{E}}^* : \mathbf{Set} \to \mathcal{E}$ of the unique geometric morphism $\gamma_{\mathcal{E}} : \mathcal{E} \to \mathbf{Set}$ is faithful if and only if $\mathcal{E}$ is non-trivial (i.e. $1_{\mathcal{E}} \not\cong 0_{\mathcal{E}}$).*

Proof It is clear that if $\mathcal{E}$ is trivial then $\gamma_{\mathcal{E}}^* : \mathbf{Set} \to \mathcal{E}$ is not faithful, it being the constant functor with value $0_{\mathcal{E}}$.

In the converse direction, suppose that $\mathcal{E}$ is non-trivial. Given two functions $f, g : A \to B$ in $\mathbf{Set}$, the arrows $\gamma_{\mathcal{E}}^*(f), \gamma_{\mathcal{E}}^*(g) : \coprod_{a \in A} 1_{\mathcal{E}} \to \coprod_{b \in B} 1_{\mathcal{E}}$ in $\mathcal{E}$ are characterized by the following identities: for any $a \in A$, $\gamma_{\mathcal{E}}^*(f) \circ s_a = t_{f(a)}$ and $\gamma_{\mathcal{E}}^*(g) \circ s_a = t_{g(a)}$, where $s_a : 1_{\mathcal{E}} \to \coprod_{a \in A} 1_{\mathcal{E}}$ and $t_b : 1_{\mathcal{E}} \to \coprod_{b \in B} 1_{\mathcal{E}}$ are respectively the a-th and b-th coproduct arrows (for any $a \in A$ and $b \in B$); so $\gamma_{\mathcal{E}}^*(f) = \gamma_{\mathcal{E}}^*(g)$ if and only if $t_{f(a)} = t_{g(a)}$ for every $a \in A$. Since distinct coproduct arrows are disjoint from each other and $\mathcal{E}$ is non-trivial, it follows that $f(a) = g(a)$ for every $a \in A$. Therefore $f = g$, as required. □

Corollary 6.3.20 *Let $\mathbb{T}$ be a geometric theory over a signature Σ, $\mathcal{E}$ a non-trivial Grothendieck topos (i.e. $1_{\mathcal{E}} \not\cong 0_{\mathcal{E}}$), M and N two $\mathbb{T}$-models in $\mathbf{Set}$ and f a function sending each sort A of Σ to a map $f_A : MA \to NA$ in such a way that the assignment $A \to \gamma_{\mathcal{E}}^*(f_A) : \gamma_{\mathcal{E}}^*(MA) \to \gamma_{\mathcal{E}}^*(NA)$ is a $\mathbb{T}$-model homomorphism in $\mathcal{E}$. Then the assignment $f : A \to f_A$ defines a $\mathbb{T}$-model homomorphism $M \to N$ in $\mathbf{Set}$.*

Proof We have to prove that f preserves the interpretation of function symbols of Σ and the satisfaction of atomic formulae over Σ. The preservation by f of the interpretation of function symbols of Σ can be expressed as the commutativity of certain squares involving finite products of sets of the form MA and NA and hence follows from that of $\gamma_{\mathcal{E}}^*(f)$ by virtue of Lemma 6.3.19. Let $R \rightarrowtail A_1 \cdots A_n$ be a relation symbol of Σ. From the fact that $\gamma_{\mathcal{E}}^*(f) : \gamma_{\mathcal{E}}^*(M) \to \gamma_{\mathcal{E}}^*(N)$ preserves the satisfaction by R it follows that, for any n-tuple $\vec{a} = (a_1, \ldots, a_n) \in MR$, the coproduct arrow: $1_{\mathcal{E}} \to \coprod_{(b_1, \ldots, b_n) \in NA_1 \times \cdots \times NA_n} 1_{\mathcal{E}}$ corresponding to the n-tuple $(f_{A_1}(a_1), \ldots, f_{A_n}(a_n))$ factors through the subobject

$\coprod_{(b_1,\ldots,b_n)\in NR} 1_{\mathcal{E}} \rightarrowtail \coprod_{(b_1,\ldots,b_n)\in NA_1\times\cdots\times NA_n} 1_{\mathcal{E}}$. But this implies, since distinct coproduct arrows are disjoint from each other and the topos $\mathcal{E}$ is non-trivial, that $f(\vec{a}) \in NR$, as required. □

Corollary 6.3.21 *Let Σ be a signature without relation symbols and $\mathbb{T}$ a geometric theory over a signature Σ' obtained from Σ by solely adding relation symbols whose interpretation in any $\mathbb{T}$-model in a Grothendieck topos is the complement of the interpretation of a geometric formula over Σ. Let $\mathcal{K}$ be a full subcategory of* f.p.$\mathbb{T}$-mod(**Set**)*, a and b be $\mathbb{T}$-models in $\mathcal{K}$, M be a $\mathbb{T}$-model in a Grothendieck topos $\mathcal{E}$ and $f : a \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, $g : b \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ be Σ'-structure homomorphisms, where E is an object of $\mathcal{E}$. Let k be an assignment sending any sort A of Σ to a function $k_A : aA \to bA$ such that $g_A \circ k_A = f_A$. Suppose that either of the following conditions is satisfied:*

- (i) *There are no relation symbols in Σ' except possibly for binary relation symbols which are $\mathbb{T}$-provable complements of the equality relation on some sort of Σ, and the Σ'-structure homomorphisms f and g are sortwise monic.*
- (ii) *$E \not\cong 0_{\mathcal{E}}$ and the $\mathbb{T}$-model homomorphism $\tilde{g} : \gamma^*_{\mathcal{E}/E}(b) \to M$ corresponding to g via the identification of Proposition 6.2.3 is sortwise monic.*

Then the assignment $k : A \to k_A$ defines a $\mathbb{T}$-model homomorphism $a \to b$ in **Set**.

Proof The assignment k preserves the interpretation of any binary relation which is $\mathbb{T}$-provably complemented to the equality relation on some sort of Σ if and only if it is monic at this sort (cf. Lemma 7.2.11 below). This holds under hypothesis (i) since f is sortwise monic and, for every sort A of Σ, $g_A \circ k_A = f_A$. On the other hand, under hypothesis (i), the fact that k preserves the interpretation of function symbols of Σ' follows from the fact that g and f do, since g is sortwise monic and $g_A \circ k_A = f_A$ for every sort A of Σ. This shows that our thesis is satisfied under hypothesis (i).

Let us now suppose that hypothesis (ii) holds. Consider the $\mathbb{T}$-model homomorphisms $\tilde{f} : \gamma^*_{\mathcal{E}/E}(a) \to M$ and $\tilde{g} : \gamma^*_{\mathcal{E}/E}(b) \to M$ corresponding to f and g via the identification of Proposition 6.2.3. Clearly, the assignment $A \to \tilde{k}_A = \gamma^*_{\mathcal{E}/E}(k_A)$ satisfies $\tilde{g}_A \circ \tilde{k}_A = \tilde{f}_A$ for all sorts A of Σ. It thus follows from Lemma 6.3.18 that $A \to \tilde{k}_A$ is a $\mathbb{T}$-model homomorphism $\gamma^*_{\mathcal{E}/E}(a) \to \gamma^*_{\mathcal{E}/E}(b)$; but this in turn implies, by Corollary 6.3.20 (notice that $E \not\cong 0_{\mathcal{E}}$ if and only if the topos $\mathcal{E}/E$ is non-trivial), that the assignment $A \to k_A$ defines a $\mathbb{T}$-model homomorphism $a \to b$, as required. □

Theorem 6.3.22 *Let Σ be a signature without relation symbols and $\mathbb{T}$ a geometric theory over a signature Σ' obtained from Σ by solely adding relation symbols whose interpretation in any $\mathbb{T}$-model in a Grothendieck topos is the complement of the interpretation of some geometric formula over Σ (for instance, $\mathbb{T}$ can be the injectivization of a geometric theory over Σ in the sense of Definition 7.2.10). Let*

$\mathcal{K}$ be a full subcategory of f.p.$\mathbb{T}$-mod(**Set**) *whose objects are all finitely generated $\mathbb{T}$-models. Suppose that either of the following conditions is satisfied:*

(i) *All the arrows in $\mathcal{K}$ are sortwise injective homomorphisms and there are no relation symbols except possibly for binary relation symbols which are $\mathbb{T}$-provable complements of the equality relation on some sort of Σ.*

(ii) *All the $\mathbb{T}$-model homomorphisms in any Grothendieck topos are sortwise monic.*

Then $\mathbb{T}$ satisfies condition (iii)(b)-(2) of Remark 6.3.2(f) with respect to the category $\mathcal{K}$.

Proof By Theorem 6.3.15, we have to verify that for any flat functor $F : \mathcal{K}^{\rm op} \to \mathcal{E}$, any $\mathbb{T}$-model c in $\mathcal{K}$, any object E of $\mathcal{E}$ and any Σ-structure homomorphism $z : c \to \mathrm{Hom}_{\mathcal{E}}(E, \tilde{F}(M_{\mathbb{T}}))$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a generalized element $x_i : E_i \to F(c)$ such that $\mathrm{Hom}_{\mathcal{E}}(e_i, \tilde{F}(M_{\mathbb{T}})) \circ z = \xi_{(c,x_i)}$.

If $E \cong 0_{\mathcal{E}}$ then the condition is trivially satisfied; indeed, one can take $I = \emptyset$. We shall therefore suppose $E \not\cong 0_{\mathcal{E}}$ or, equivalently, that the topos $\mathcal{E}/E$ is non-trivial.

In this proof, we shall suppose for simplicity that Σ is one-sorted, but all our arguments can be readily extended to the general case.

Let $\{r_1, \ldots, r_n\}$ be a set of generators for the model c. Consider their images $z(r_1), \ldots, z(r_n) : E \to \tilde{F}(M_{\mathbb{T}})$ under the homomorphism z. By Proposition 5.2.11, for any $i \in \{1, \ldots, n\}$ there exist an epimorphic family $\{e^i_j : E^i_j \to E \mid j \in I_i\}$ in $\mathcal{E}$ and for each $j \in J_i$ a $\mathbb{T}$-model a^i_j in $\mathcal{K}$, a generalized element $x^i_j : E^i_j \to F(a^i_j)$ and an element $y^i_j \in a^i_j$ such that $\xi_{(a^i_j, x^i_j)}(y^i_j) = z(r_i) \circ e^i_j$.

For any tuple $\vec{k} = (k_1, \ldots, k_n) \in J_1 \times \cdots \times J_n$, consider the iterated pullback $e_{\vec{k}} : E_{\vec{k}} := E^1_{k_1} \times_E \cdots \times_E E^n_{k_n} \to E$. The family of arrows $\{e_{\vec{k}} : E_{\vec{k}} \to E \mid \vec{k} \in J_1 \times \cdots \times J_n\}$ is clearly epimorphic. For any $i \in \{1, \ldots, n\}$ and $k_i \in J_i$, set $x^i_{\vec{k}} : E_{\vec{k}} \to F(a^i_{k_i})$ equal to the composite of the generalized element $x^i_{k_i} : E^i_{k_i} \to F(a^i_{k_i})$ with the canonical pullback arrow $p^i_{k_i} : E_{\vec{k}} \to E^i_{k_i}$. For any fixed $\vec{k} \in J_1 \times \cdots \times J_n$, by inductively applying the 'joint embedding property' of Proposition 5.2.11 we can find an epimorphic family $\{u^{\vec{k}}_l : U^{\vec{k}}_l \to E_{\vec{k}} \mid l \in L_{\vec{k}}\}$ and for each $l \in L_{\vec{k}}$ a $\mathbb{T}$-model $d^{\vec{k}}_l$ in $\mathcal{K}$, a generalized element $w^{\vec{k}}_l : U^{\vec{k}}_l \to F(d^{\vec{k}}_l)$ and arrows $f^{\vec{k}}_{i,l} : a^i_{k_i} \to d^{\vec{k}}_l$ in $\mathcal{K}$ such that $x^i_{\vec{k}} \circ u^{\vec{k}}_l = F(f^{\vec{k}}_{i,l}) \circ w^{\vec{k}}_l$. Let us prove that $z(r_i) \circ e_{\vec{k}} \circ u^{\vec{k}}_l = \xi_{(d^{\vec{k}}_l, w^{\vec{k}}_l)}(f^{\vec{k}}_{i,l}(y^i_{k_i}))$ (for any $i, \vec{k}$ and l). We have already observed that for any $i \in \{1, \ldots, n\}$, we have $z(r_i) \circ e^i_{k_i} = \xi_{(a^i_{k_i}, x^i_{k_i})}(y^i_{k_i})$. Composing both terms of this equation with $p^i_{k_i}$ and applying Lemma 5.2.10(ii) yields the equality $z(r_i) \circ e_{\vec{k}} = \xi_{(a^i_{k_i}, x^i_{\vec{k}})}(y^i_{k_i})$. On the other hand, by Lemma 5.2.10(i) we have that $\xi_{(a^i_{k_i}, x^i_{\vec{k}})}(y^i_{k_i}) \circ u^{\vec{k}}_l = \xi_{(d^{\vec{k}}_l, w^{\vec{k}}_l)}(f^{\vec{k}}_{i,l}(y^i_{k_i}))$. Our claim thus follows by combining the former identity with the one obtained by composing both sides of the latter identity with $u_l^{\vec{k}}$.

Let us now consider the Σ'-structure homomorphisms

$$\mathrm{Hom}_{\mathcal{E}}(e_{\vec{k}} \circ u_l^{\vec{k}}, \tilde{F}(M_{\mathbb{T}})) \circ z : c \to \mathrm{Hom}_{\mathcal{E}}(U_l^{\vec{k}}, \tilde{F}(M_{\mathbb{T}}))$$

and

$$\xi_{(d_l^{\vec{k}}, w_l^{\vec{k}})} : d_l^{\vec{k}} \to \mathrm{Hom}_{\mathcal{E}}(U_l^{\vec{k}}, \tilde{F}(M_{\mathbb{T}})).$$

Since, under either hypothesis (i) or (ii), the arrows of the category $\mathcal{K}$ are sortwise monic, by Proposition 6.3.16 the homomorphism $\xi_{(d_l^{\vec{k}}, w_l^{\vec{k}})}$ is injective. Therefore, since the image of all the generators $r_1, \ldots, r_n$ of c under the homomorphism $\mathrm{Hom}_{\mathcal{E}}(e_{\vec{k}} \circ u_l^{\vec{k}}, \tilde{F}(M_{\mathbb{T}})) \circ z$ is contained in the image of the homomorphism $\xi_{(d_l^{\vec{k}}, w_l^{\vec{k}})}$, an easy induction on the structure $t(r_1, \ldots, r_n)$ of the elements of c shows that the image of every element of c belongs to the image of $\xi_{(d_l^{\vec{k}}, w_l^{\vec{k}})}$; in other words, there exists a factorization $n_l^{\vec{k}} : c \to d_l^{\vec{k}}$ of $\mathrm{Hom}_{\mathcal{E}}(e_{\vec{k}} \circ u_l^{\vec{k}}, \tilde{F}(M_{\mathbb{T}})) \circ z$ across $\xi_{(d_l^{\vec{k}}, w_l^{\vec{k}})}$. By Corollary 6.3.21, this factorization is a $\mathbb{T}$-model homomorphism. We have thus found an epimorphic family on E, namely $\{e_{\vec{k}} \circ u_l^{\vec{k}} \mid \vec{k} \in J_1 \times \cdots \times J_n,\ l \in L_{\vec{k}}\}$, and for every $\vec{k}$ and l a $\mathbb{T}$-model homomorphism $n_l^{\vec{k}} : c \to d_l^{\vec{k}}$ in $\mathcal{K}$ such that

$$\mathrm{Hom}_{\mathcal{E}}(e_{\vec{k}} \circ u_l^{\vec{k}}, \tilde{F}(M_{\mathbb{T}})) \circ z = \xi_{(d_l^{\vec{k}}, w_l^{\vec{k}})} \circ n_l^{\vec{k}}.$$

The composites $v_l^{\vec{k}} : U_l^{\vec{k}} \to F(c)$ of the generalized elements $w_l^{\vec{k}} : U_l^{\vec{k}} \to F(d_l^{\vec{k}})$ with the arrows $F(n_l^{\vec{k}})$ thus yield generalized elements such that $\mathrm{Hom}_{\mathcal{E}}(e_{\vec{k}} \circ u_l^{\vec{k}}, \tilde{F}(M_{\mathbb{T}})) \circ z = \xi_{(d_l^{\vec{k}}, w_l^{\vec{k}})} \circ n_l^{\vec{k}} = \xi_{(c, v_l^{\vec{k}})}$, where the last equality follows by Lemma 5.2.10(i). This completes our proof. □

Remark 6.3.23 The assumption in condition (i) of the theorem that all the arrows in $\mathcal{K}$ be sortwise monic is weaker than the requirement of condition (ii) that all the $\mathbb{T}$-model homomorphisms in any Grothendieck topos be sortwise monic. Anyway, condition (ii) is necessary for $\mathbb{T}$ to be classified by the topos $[\mathcal{K}, \mathbf{Set}]$ if all the homomorphisms in $\mathcal{K}$ are sortwise injective (cf. Corollary 7.2.9 below).

Corollary 6.3.24 *Let $\mathbb{T}$ be a geometric theory whose model homomorphisms in any Grothendieck topos are monic and which satisfies the hypotheses of Theorem 6.3.22 with respect to a full subcategory $\mathcal{K}$ of* f.p.$\mathbb{T}$-mod($\mathbf{Set}$). *Then $\mathbb{T}$ is of presheaf type classified by the topos $[\mathcal{K}, \mathbf{Set}]$ if and only if it satisfies condition (i) of Theorem 6.3.1 and condition (ii)(a) of Theorem 6.3.8 with respect to the category $\mathcal{K}$.*

Proof The theory $\mathbb{T}$ satisfies condition (iii) of Theorem 6.3.1 with respect to the category $\mathcal{K}$ since condition (iii)(b)-(1) of Remark 6.3.2(f) holds by Corollary

6.3.17 and condition (iii)(b)-(2) of Remark 6.3.2(f) holds by Theorem 6.3.22. By Remark 6.3.9(a) and Theorem 6.3.4, if $\mathbb{T}$ satisfies condition (i) of Theorem 6.3.1 with respect to $\mathcal{K}$ then it satisfies condition (ii)(b) of Theorem 6.3.8 whence by Theorem 6.3.8 it satisfies condition (ii) of Theorem 6.3.1 if and only if it satisfies condition (ii)(a) of Theorem 6.3.8. □

6.3.3 *Abstract reformulation*

Thanks to the general theory of indexed filtered colimits developed in section 5.1, we can reformulate the conditions of Theorem 6.3.1 in more abstract, though less explicit terms, as follows.

Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\mathcal{K}$ a full subcategory of f.p.$\mathbb{T}$-mod(**Set**).

Condition (i) of Theorem 6.3.1 for $\mathbb{T}$ with respect to $\mathcal{K}$ can be reformulated as the requirement that for every $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$, the $\mathcal{E}$-indexed category of elements $\underline{\int \overline{H_M}^{\mathrm{f}}}_{\mathcal{E}}$ of the functor H_M of Theorem 6.3.1 (or equivalently its $\mathcal{E}$-final subcategory $\underline{\int H_M}_{\mathcal{E}}$, cf. Theorem 5.1.12) be $\mathcal{E}$-filtered.

By the discussion at the end of section 5.2.3, condition (ii) of Theorem 6.3.1 can be reformulated as the requirement that for any $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ such that the functor H_M is flat, the canonical $\mathcal{E}$-indexed cocone with vertex M over the $\mathcal{E}$-indexed functor $P \circ \pi^{\mathrm{f}}_{\overline{H_M}}$ (where P is the $\mathcal{E}$-indexed functor defined at the end of section 5.2.3), or equivalently over its restriction to the $\mathcal{E}$-strictly final subcategory $\underline{\int H_M}_{\mathcal{E}}$ of $\underline{\int \overline{H_M}^{\mathrm{f}}}_{\mathcal{E}}$ (cf. Theorem 5.1.12), be colimiting.

Conditions (i) and (ii) of Theorem 6.3.1 can thus be interpreted by saying that every $\mathbb{T}$-model in a Grothendieck topos $\mathcal{E}$ is canonically an $\mathcal{E}$-filtered colimit of 'constant' $\mathbb{T}$-models coming from $\mathcal{K}$.

Under the assumption that conditions (i) and (ii) are satisfied, condition (iii) of Theorem 6.3.1 is equivalent to the requirement that for any $\mathbb{T}$-model c in $\mathcal{K}$ and any Grothendieck topos $\mathcal{E}$, the $\mathbb{T}$-model $\gamma^*_{\mathcal{E}}(c)$ be $\mathcal{E}$-finitely presentable (in the sense of section 6.2.3). Indeed, the fact that this condition is necessary for $\mathbb{T}$ to be of presheaf type follows from Proposition 6.2.13, while the fact that it is sufficient, together with conditions (i) and (ii) of Theorem 6.3.1, for $\mathbb{T}$ to be classified by the topos $[\mathcal{K}, \mathbf{Set}]$ can be proved by showing that it implies condition (iii)(a) of Theorem 6.3.1. In fact, this immediately follows from the fact that inverse image functors of geometric morphisms preserve internal colimits in view of the representation of every $\mathbb{T}$-model M in a Grothendieck topos $\mathcal{E}$ as an $\mathcal{E}$-filtered colimit of 'constant' $\mathbb{T}$-models coming from $\mathcal{K}$ provided by condition (ii).

We can thus conclude that $\mathbb{T}$ is of presheaf type classified by the topos $[\mathcal{K}, \mathbf{Set}]$ if and only if every $\mathbb{T}$-model in any Grothendieck topos $\mathcal{E}$ is an $\mathcal{E}$-filtered colimit of its canonical diagram made of 'constant' models coming from $\mathcal{K}$ and these models are $\mathcal{E}$-finitely presentable.